

ST:TNG Warp Core



Printed Parts:

- WarpCoreBase.stl
- SupportRingBase.stl
- ChamberEnd.stl (x2)
- ChamberMiddle.stl
- SupportRingToRingLower.stl
- SupportRingToRingUpper.stl
- SupportRingToMI.stl
- MatterInjector.stl
- Knob.stl
- LightRing.stl (x15)
- LightShelf.stl (x15)
- Upright.stl (x27)
- LightRingSupportUnit.stl (x9)
- Warp_Core_Dilithium_Plug.stl
- Warp_Core_MARC_Lights_Extended.stl (x2)

Hardware:

- M3 x 10mm screws
- ¼" Threaded Rod – 18"+
- ¼" Nut (x2)
- ¼" Fender Washer
- Wire (6 and 4 conductor cables will be helpful)
- LED – Blue (x120)
- LED – White (x12)
- Arduino Nano
- 2N7002 (SMC) or 2N7000 Mosfet (x11)
- 10K Resistor – SMC or Through Hole (x11)
- 10K Potentiometer
- 6 Pin Molex Connector (x3 – Digikey WM4204-ND)
- 6 Pin Molex Housing (x3 - Digikey WM2004-ND)
- 4 Pin Molex Connector (Digikey WM4202-ND)
- 4 Pin Molex Housing (Digikey WM2002-ND)
- 2 Pin Molex Connector (Digikey WM4200-ND)
- 2 Pin Molex Housing (Digikey WM2000-ND)
- Crimp Connectors Female (x24– Digikey WM1114-ND)
- Power Connector (Digikey CP-011A-ND)
- WarpCore PDB (OSHPark - https://oshpark.com/shared_projects/OjKfgvHc) or prototype board.

- Optional – Single row headers, male and female, for Nano
- Hot Glue
- 12v Power Supply

Assembly

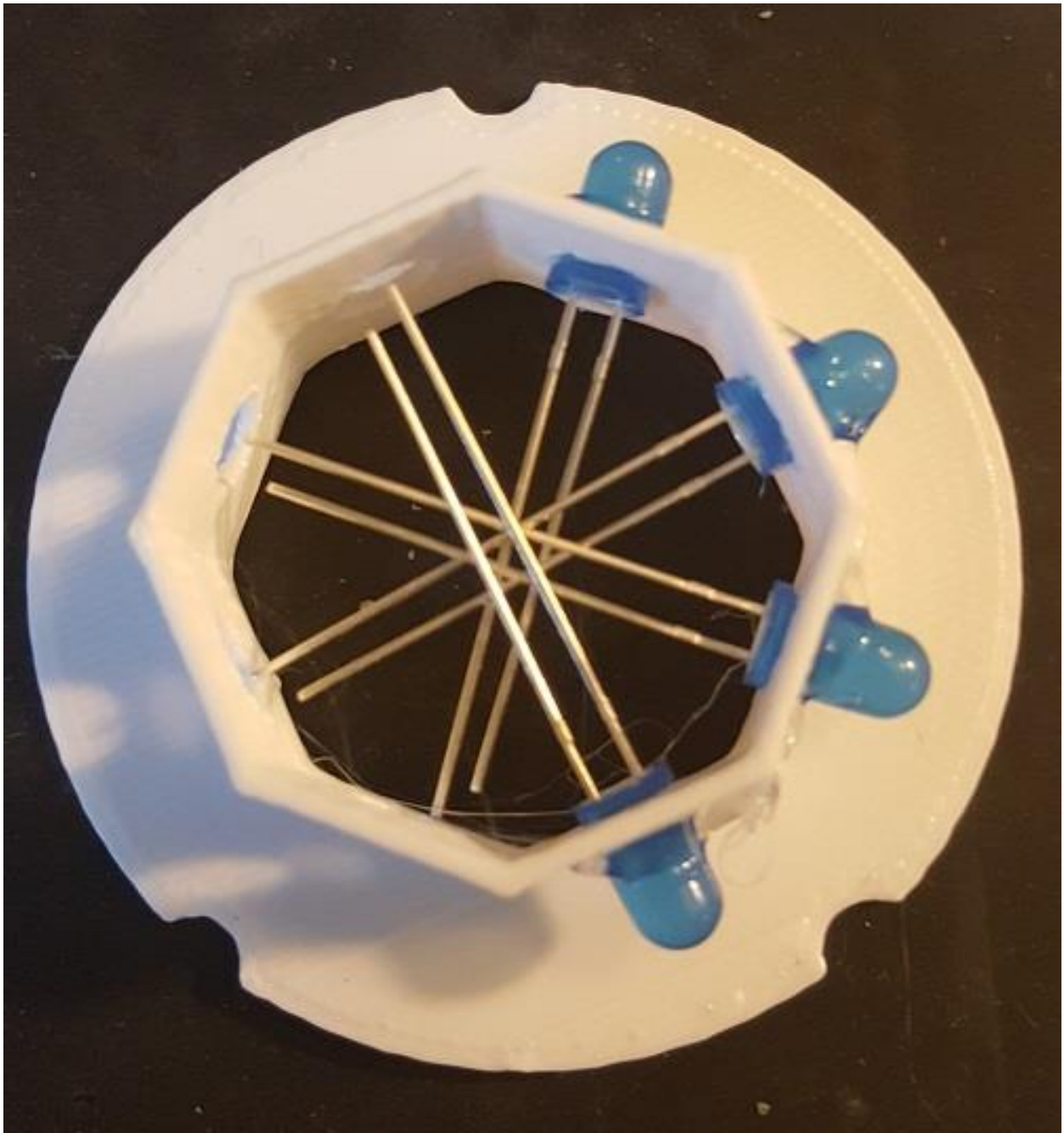
These instructions are not a complete step by step guide on assembling the Warp Core. It is a guide on how the parts fit together. While I try to be as clear as possible in describing each step, it is up to the user to determine how best to complete each step. I have included pictures to help demonstrate what things should look like that is described in the text.

I am a Software Engineer, not an English Major or manual writer. I will be the first to admit that my writing is not the best. I have tried to do the best I can to describe how to assemble things. Successful assembly will rely on the users mechanical and electronic skills. This is not a novice level project.

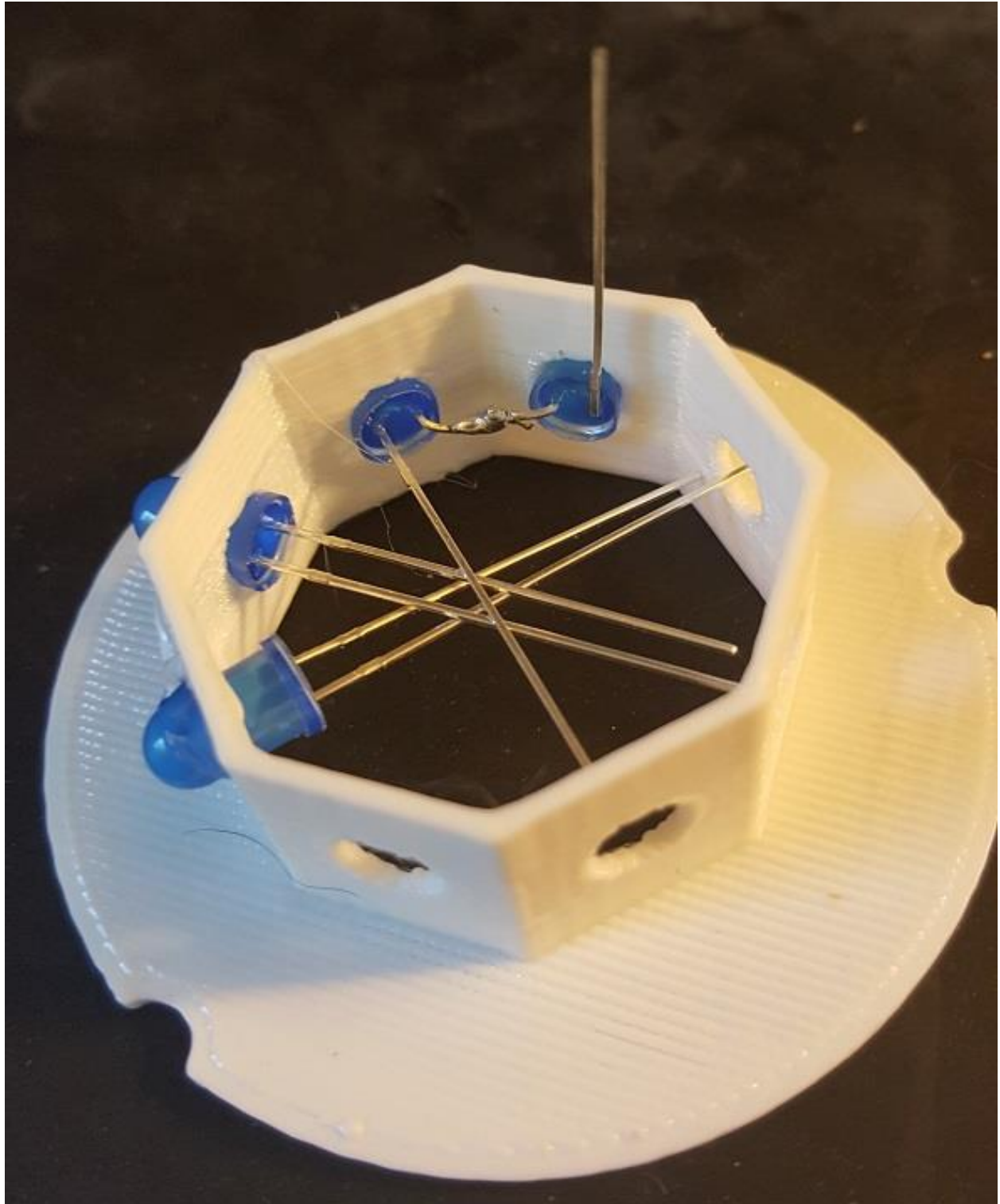
Begin by printing the light shelves and light rings. If you do them in groups of 5, you can be working on one set while the next set prints.

Each light shelf contains 8 blue LEDs connected as 2 parallel runs of 4 LEDs in series. Insert 4 LEDs, making sure the leads are orientated the same. The longer lead is the anode that connects to supply voltage. I orientated the LEDs as shown below. The anode, when looking at the picture, is the lead on

the left. Hot glue the LEDs in place.

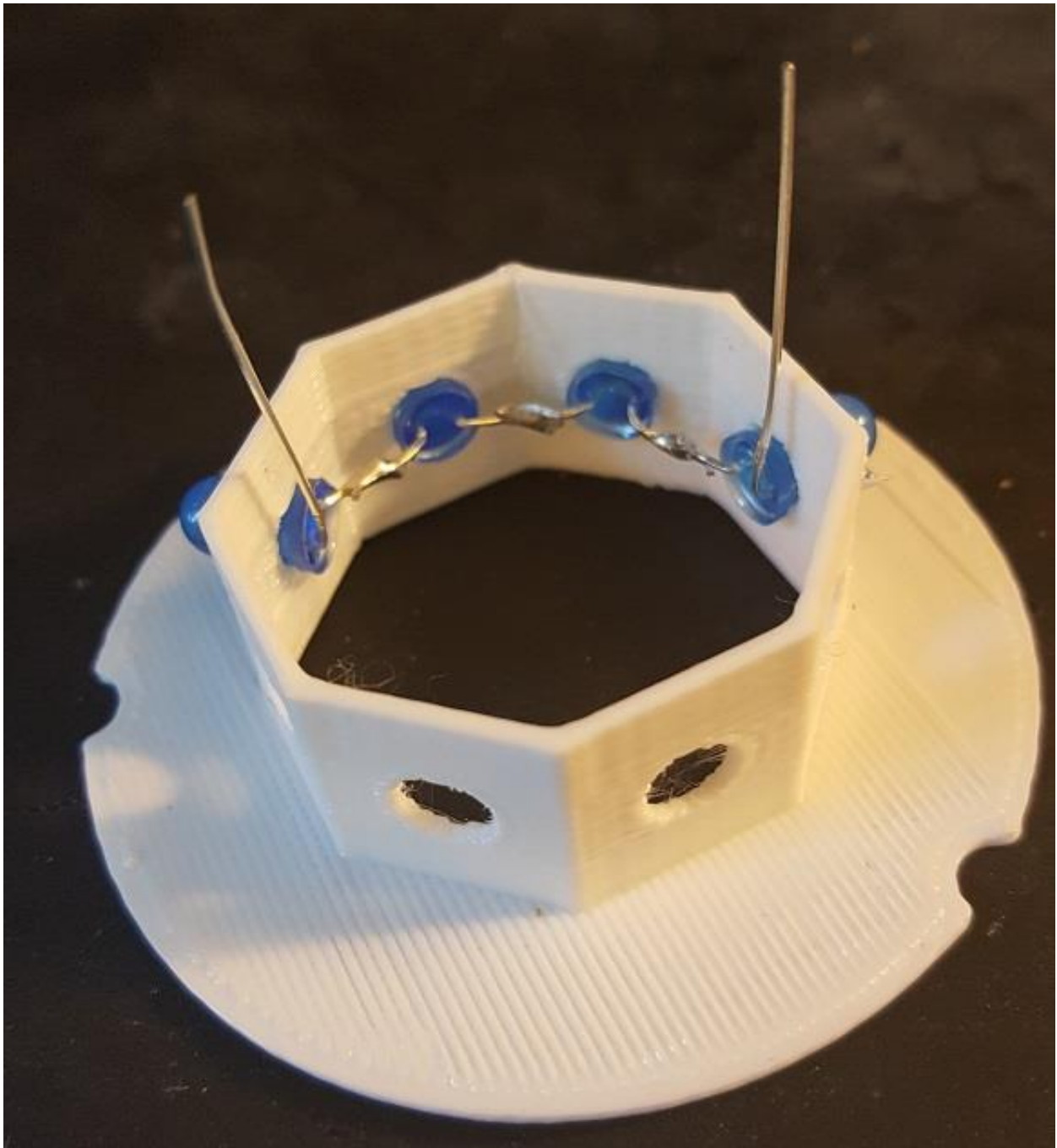


Repeat this for the other 4 shelves. This will allow time for the glue to dry before moving to the next step.



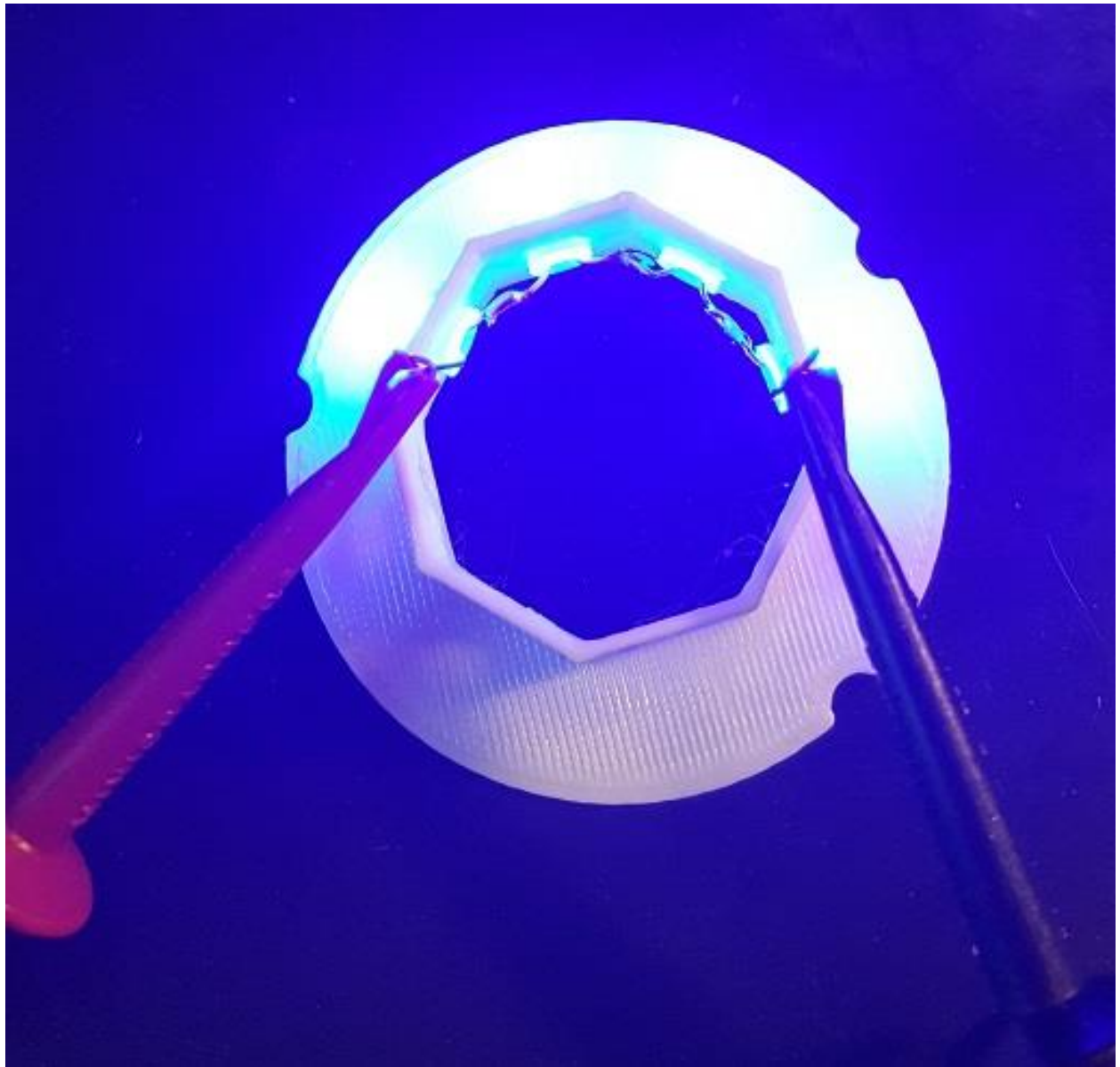
Bend the anode of the first LED up and out of the way. Connect the cathode (short lead) of the LED to the anode of the next LED. Repeat for the other LEDs. You should end up with something like the

picture below.

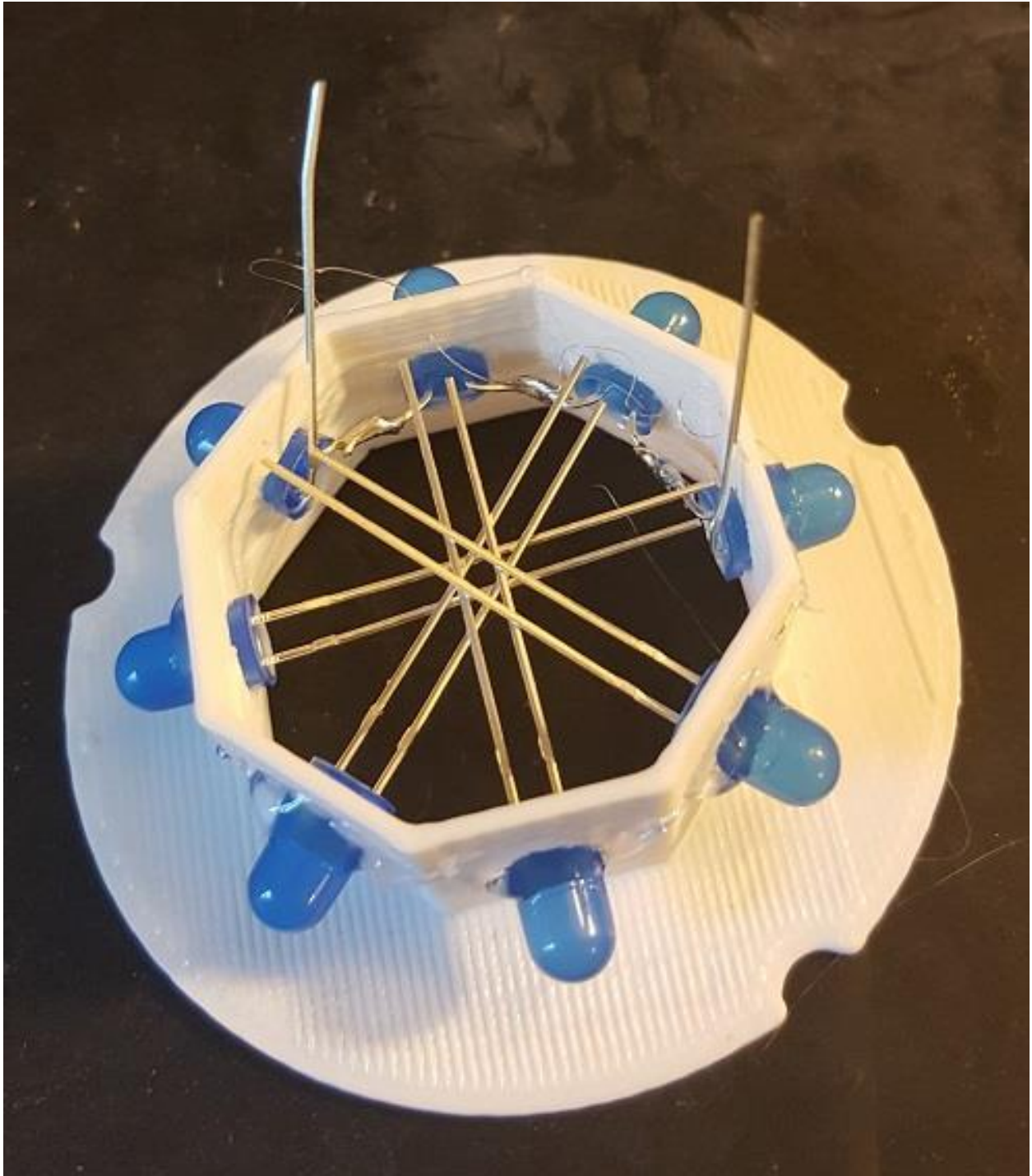


Using the 12v power supply, or a bench power supply, connect the anode (long lead) to +12vdc and the cathode (short lead) to ground, or negative, or zero, or whatever you want to call it. Make sure all the

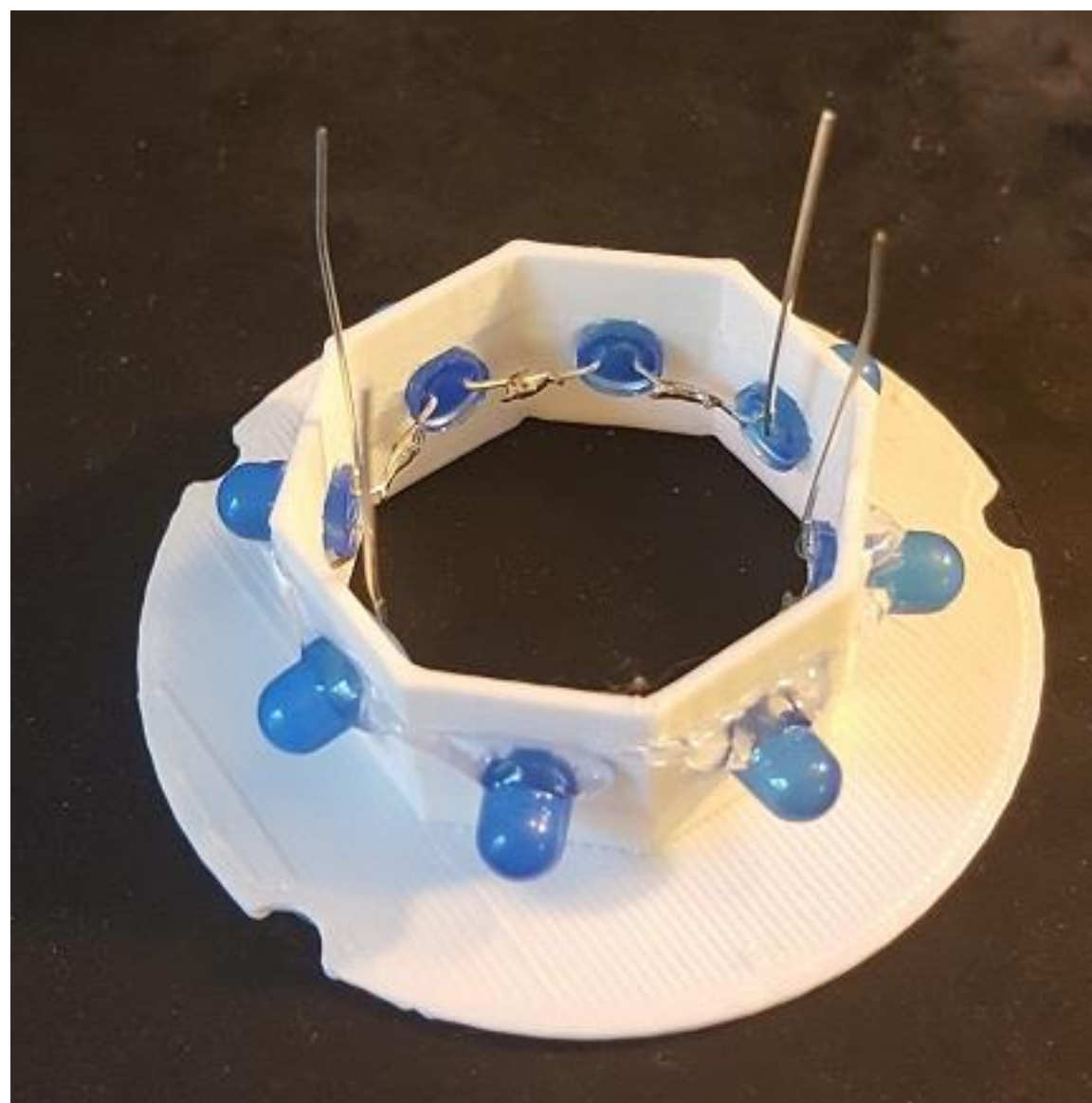
LEDs light.

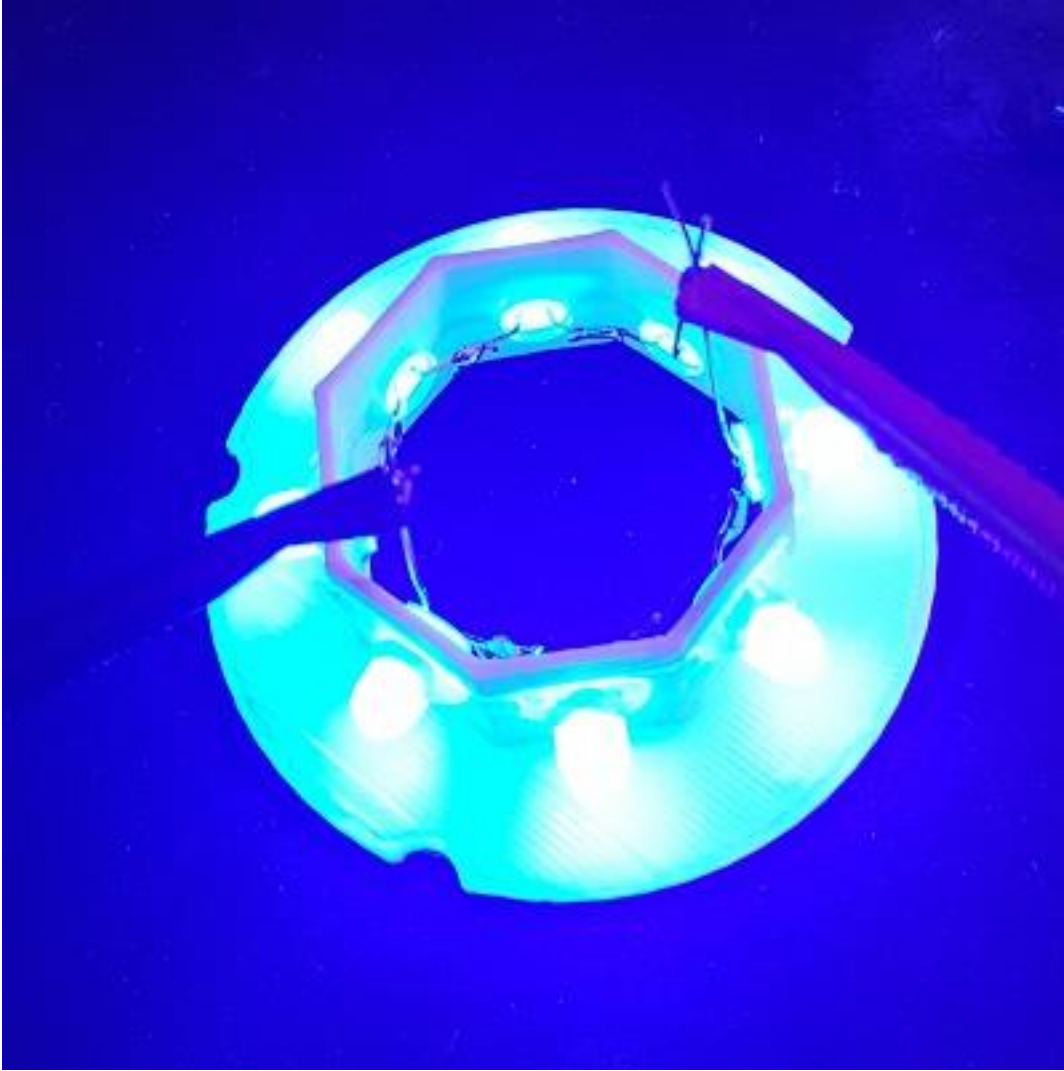


Add the other 4 LEDs following the same procedure used for the first four and hot glue the LEDs in place. Set aside so the glue and set and repeat for the other 4 shelves.

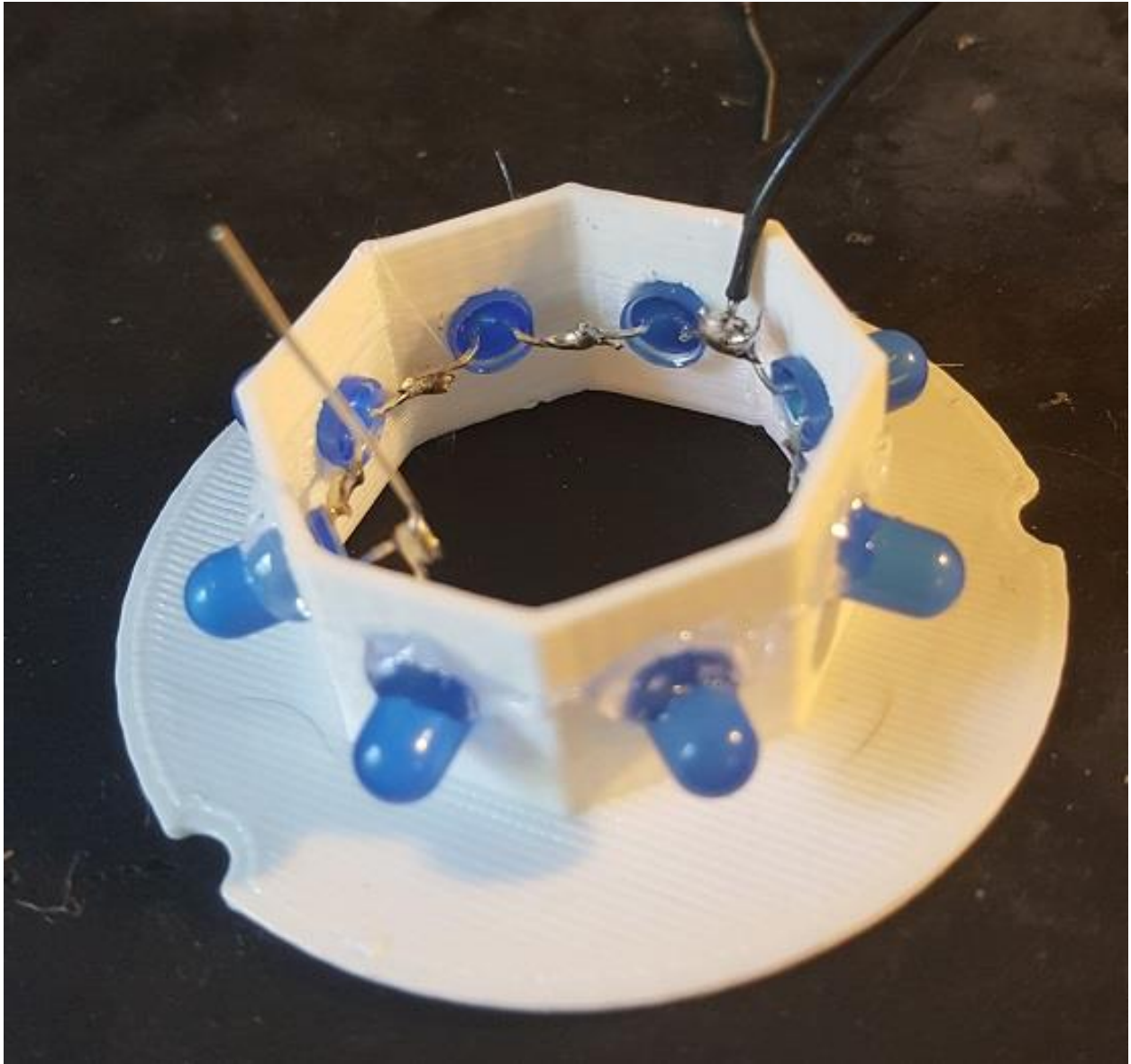


After you have all 5 shelves finished, the other 4 LEDs can be wired together in the same fashion as the first four. Using the power supply, make sure all the LEDs work.



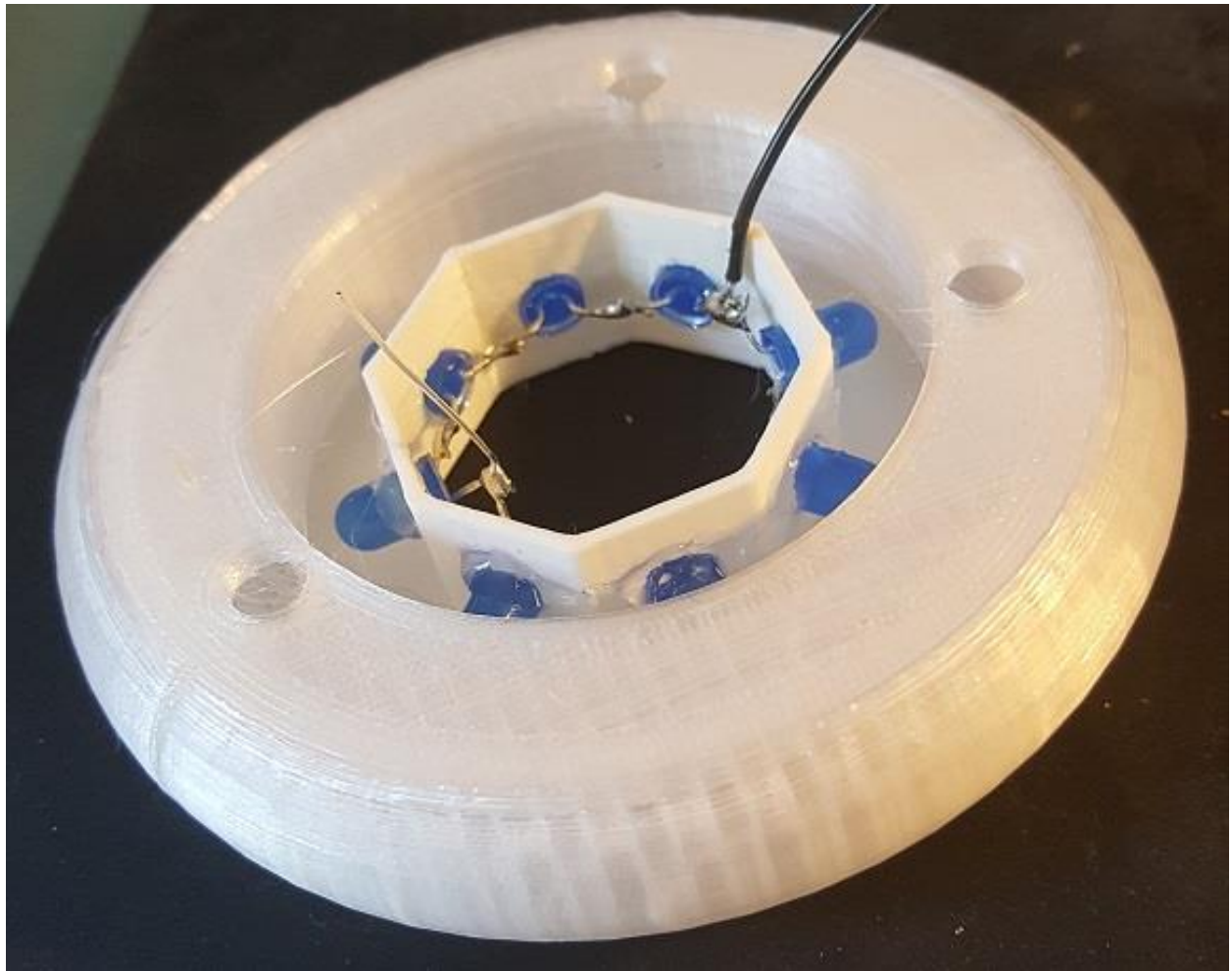


Connect the cathodes together to a wire as shown below. It is best to use various color wires to be able to identify which wire goes to each shelf. I used the wires in the following order: Black, Blue, White, Brown, Green, and Red. By using the same order on each set of shelves, it will make wiring to the Nano easier.

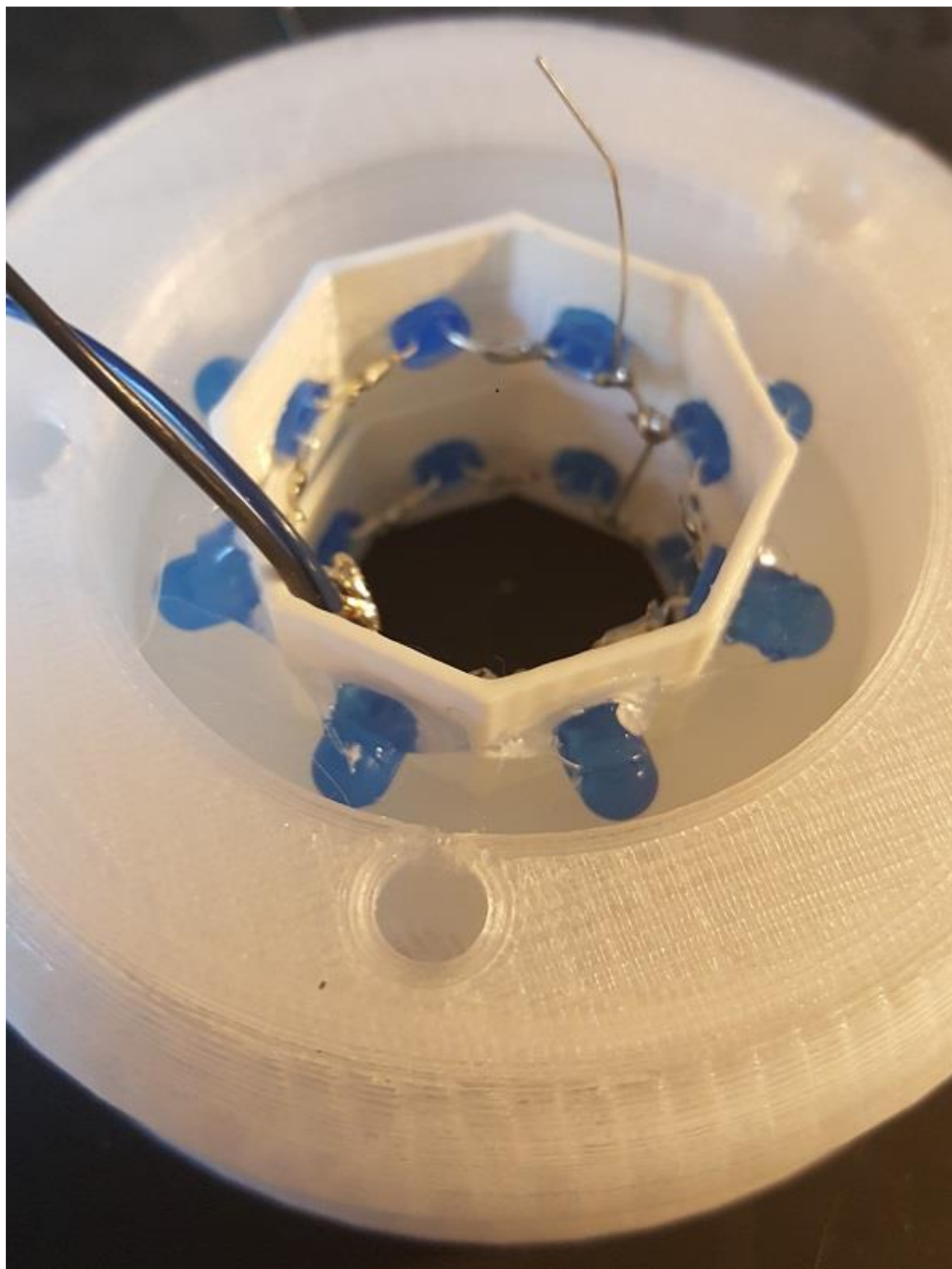


The anodes are connected together and one of the leads is left long. This will be used to connect to the next shelf.

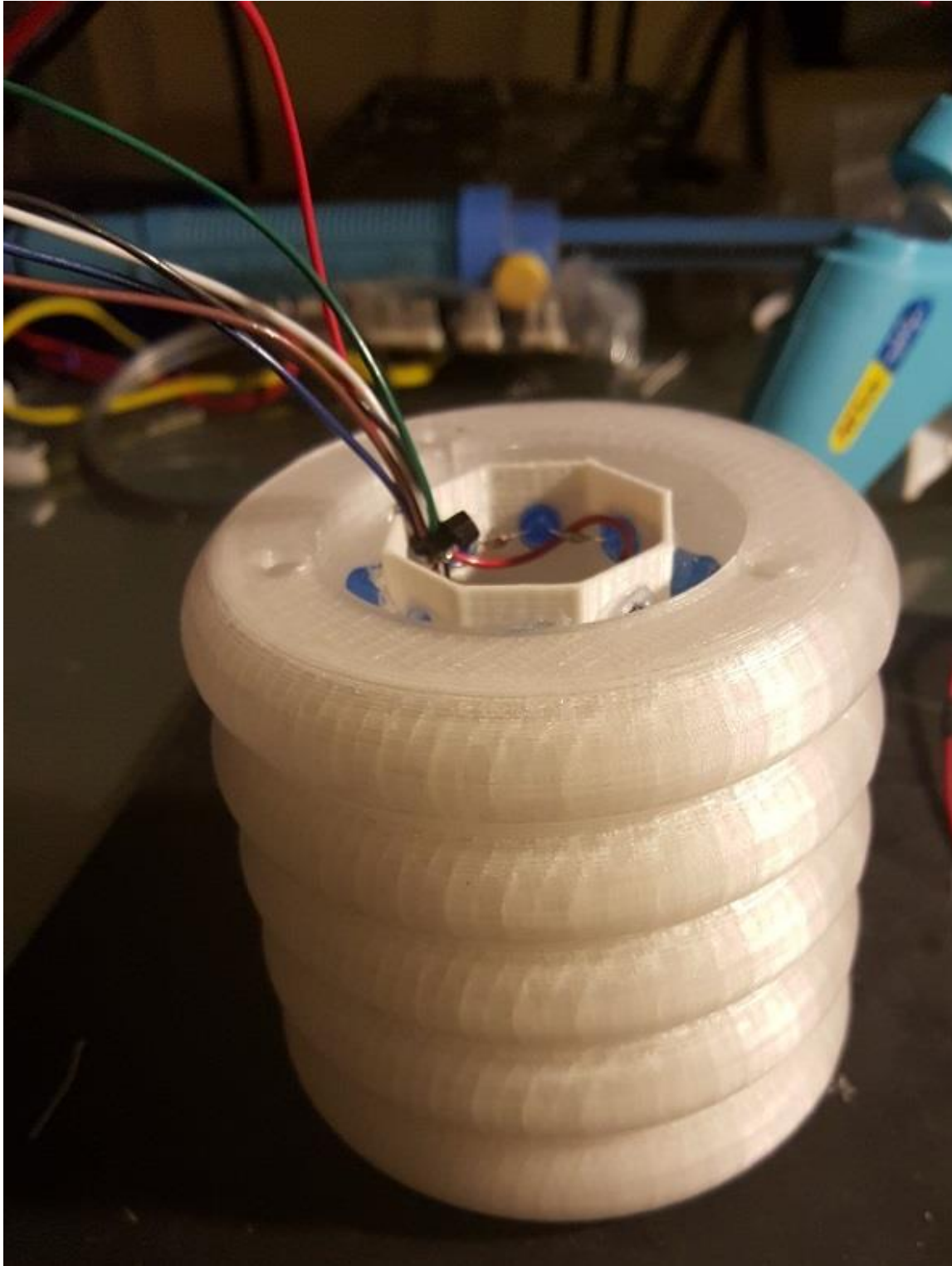
Place the light shelf into a light ring. Congratulations. You now have one done. Only 14 more to go.



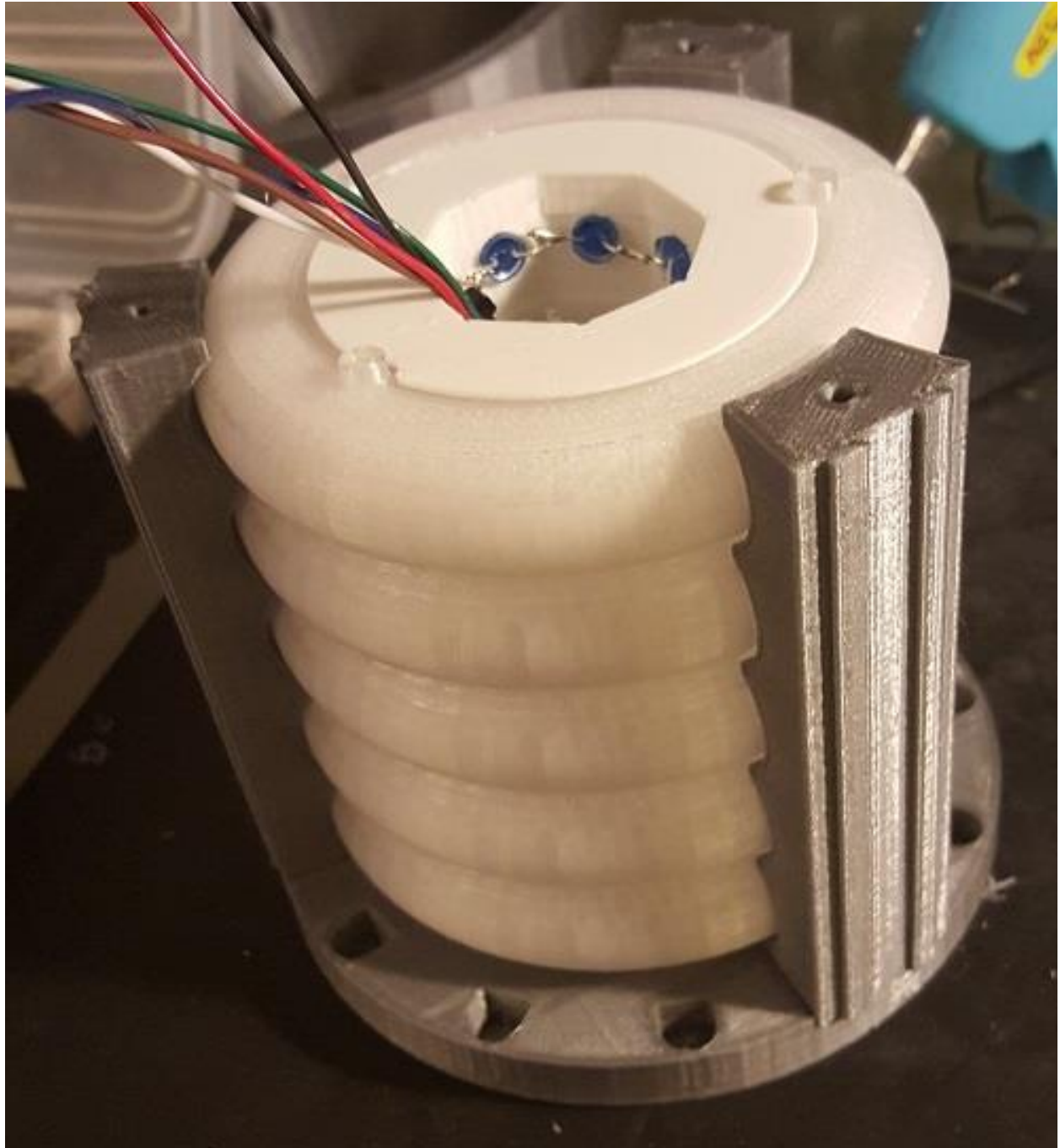
Repeat this process for the next four shelves, stacking them on top of each other. I ran the wire from the shelf below under the wire on the one above to keep the wires together. I wrapped the anode lead from the lower shelf around the anode wires of the shelf above as shown below.



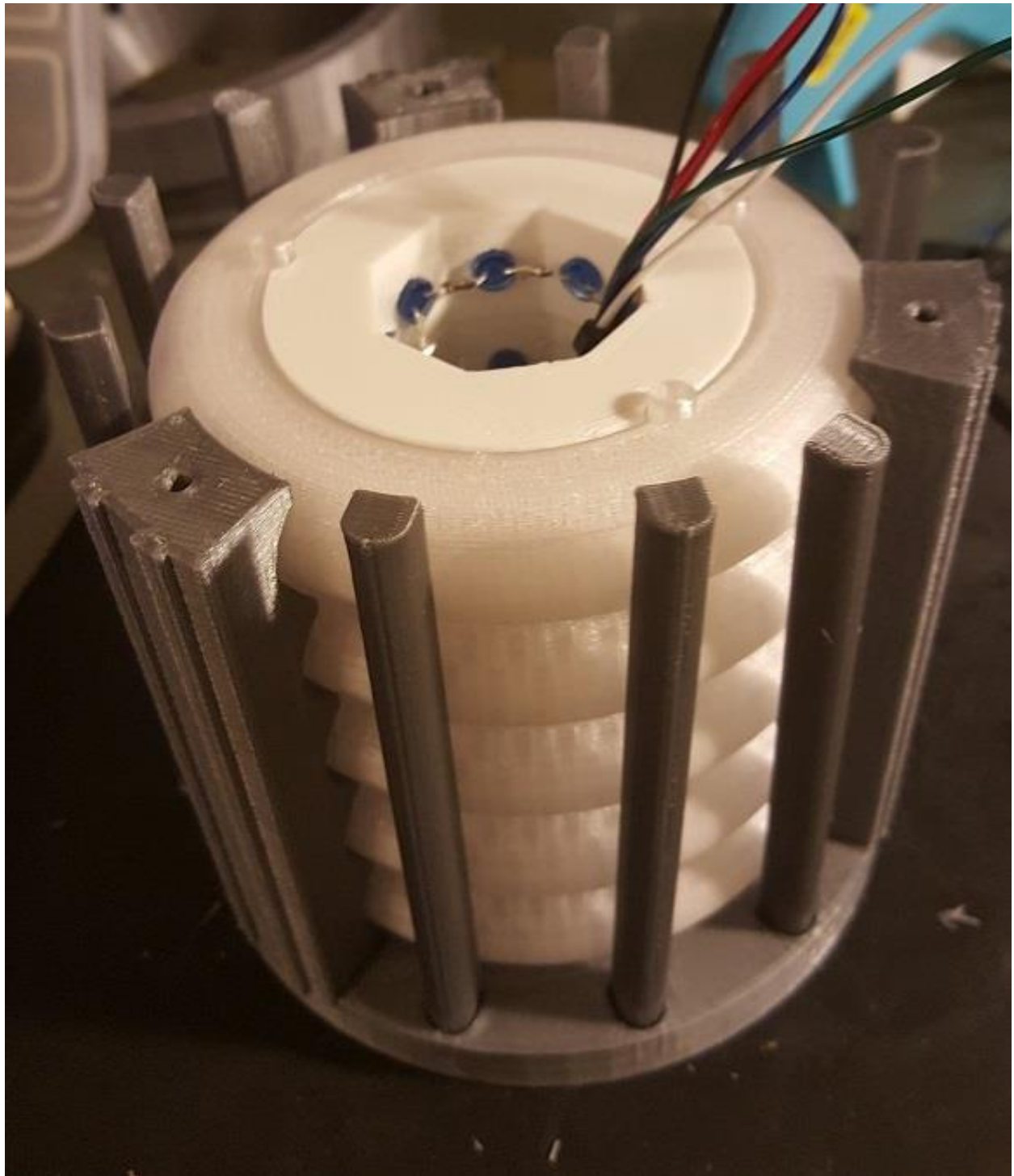
Once you have all the shelves wired together, you should have a stack of 5 rings. On the top ring, run the red wire around to the anode wires. You should have 6 wires. If you connect the Red wire to +12, you can check each ring by connecting one of the other wires to ground. Complete the other two light stacks



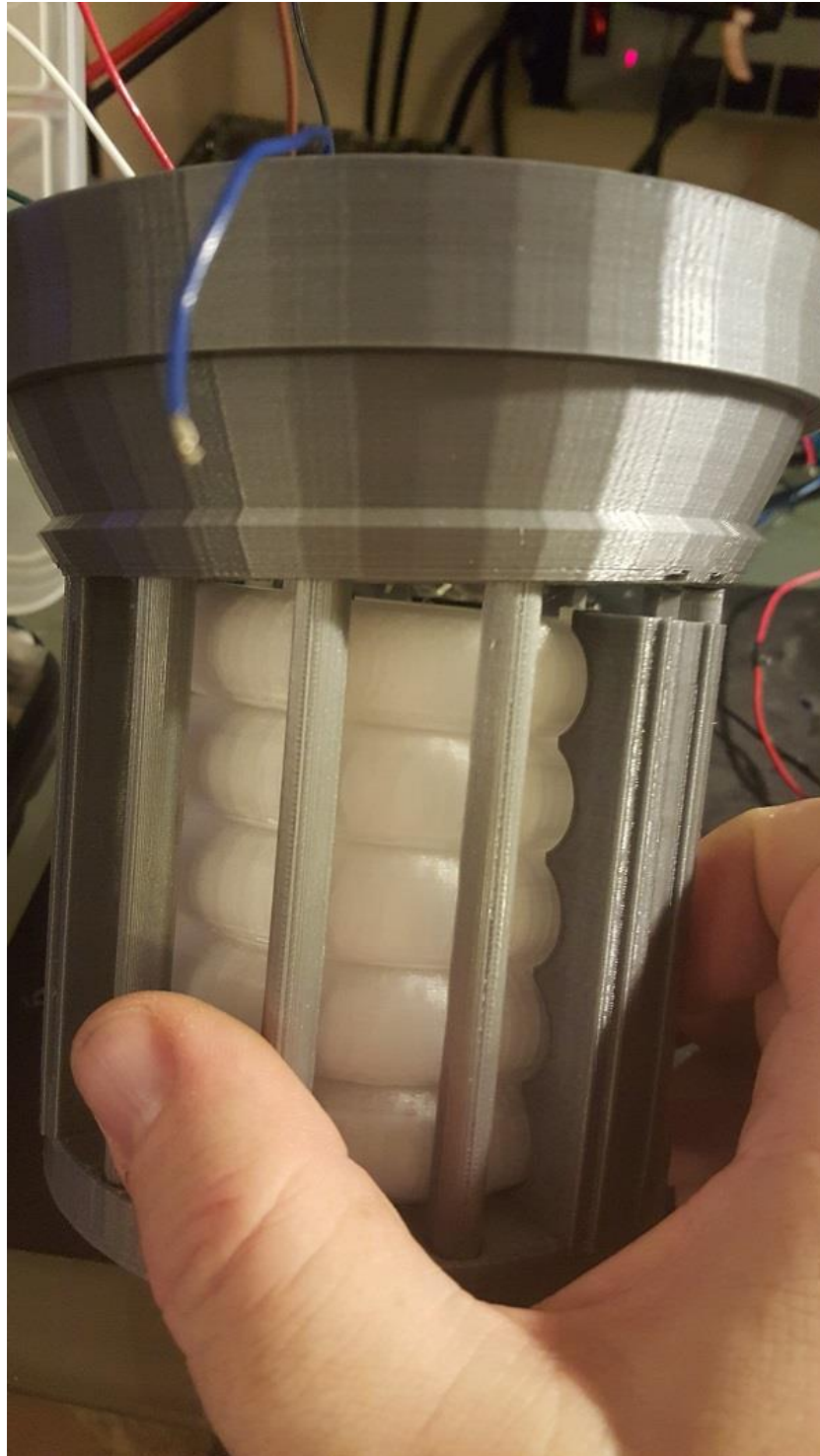
Once all 3 light stacks are finished, the three light sections can be made. Use the SupportRingBase, 3 LightRingSupportUnits, 9 Uprights, and a ChamberEnd. Place the light stack on the SupportRingBase and fit a LightRingSupportUnit next to it. Use a M3x10mm screw to hold the support unit in place. Repeat with the other 2 support units.

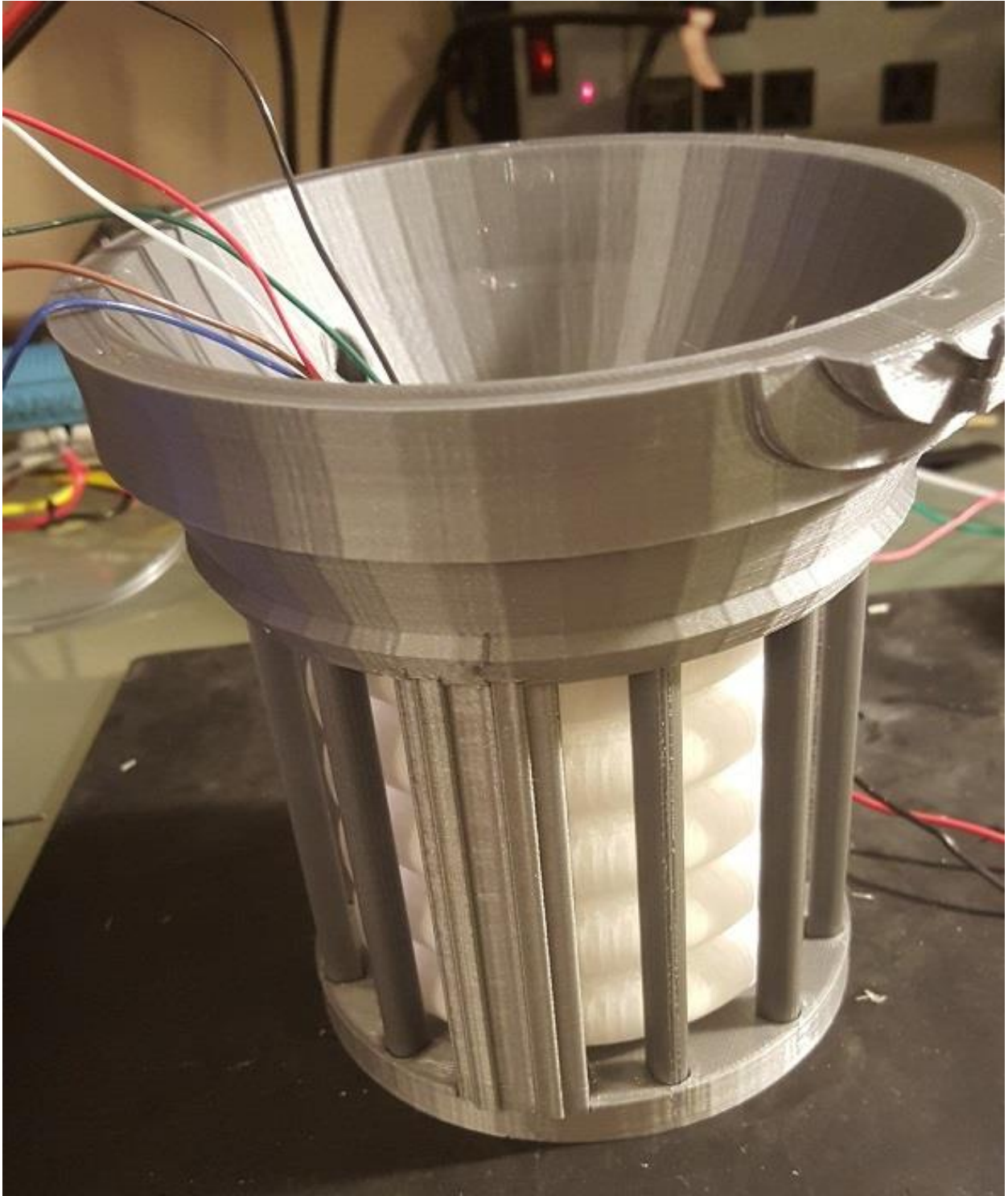


Insert the 9 uprights into the holes.



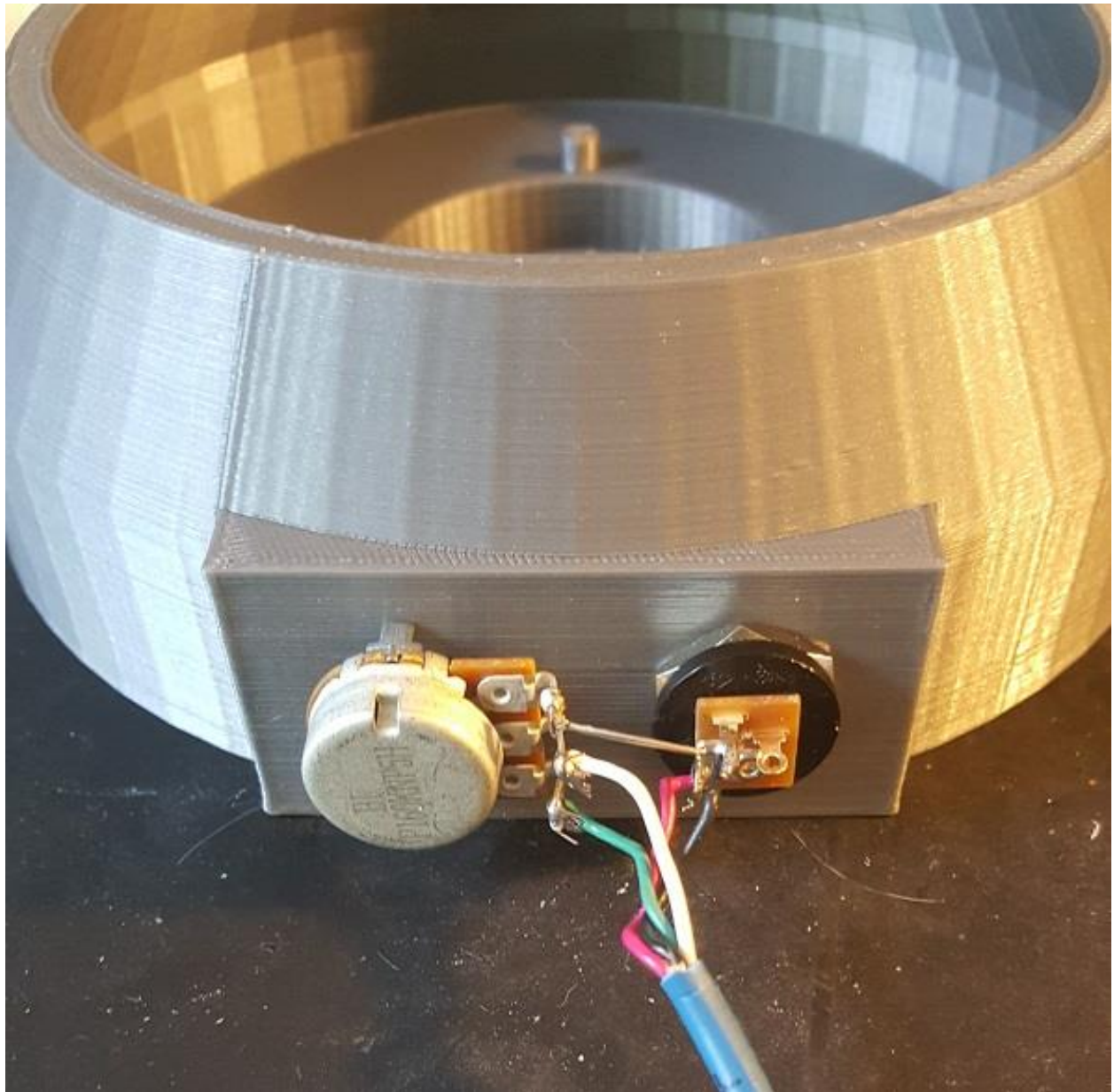
Place the ChamberEnd on top, aligning the uprights and supports into the corresponding holes. This can be a bit tricky. What I found works best was to get one support in place and the uprights next to it in place and loosely put in one screw to hold everything together. I could then get the other uprights and supports in place and put the other two screws in place and tighten them up. **DO NOT OVER TIGHTEN THE SCREWS!**





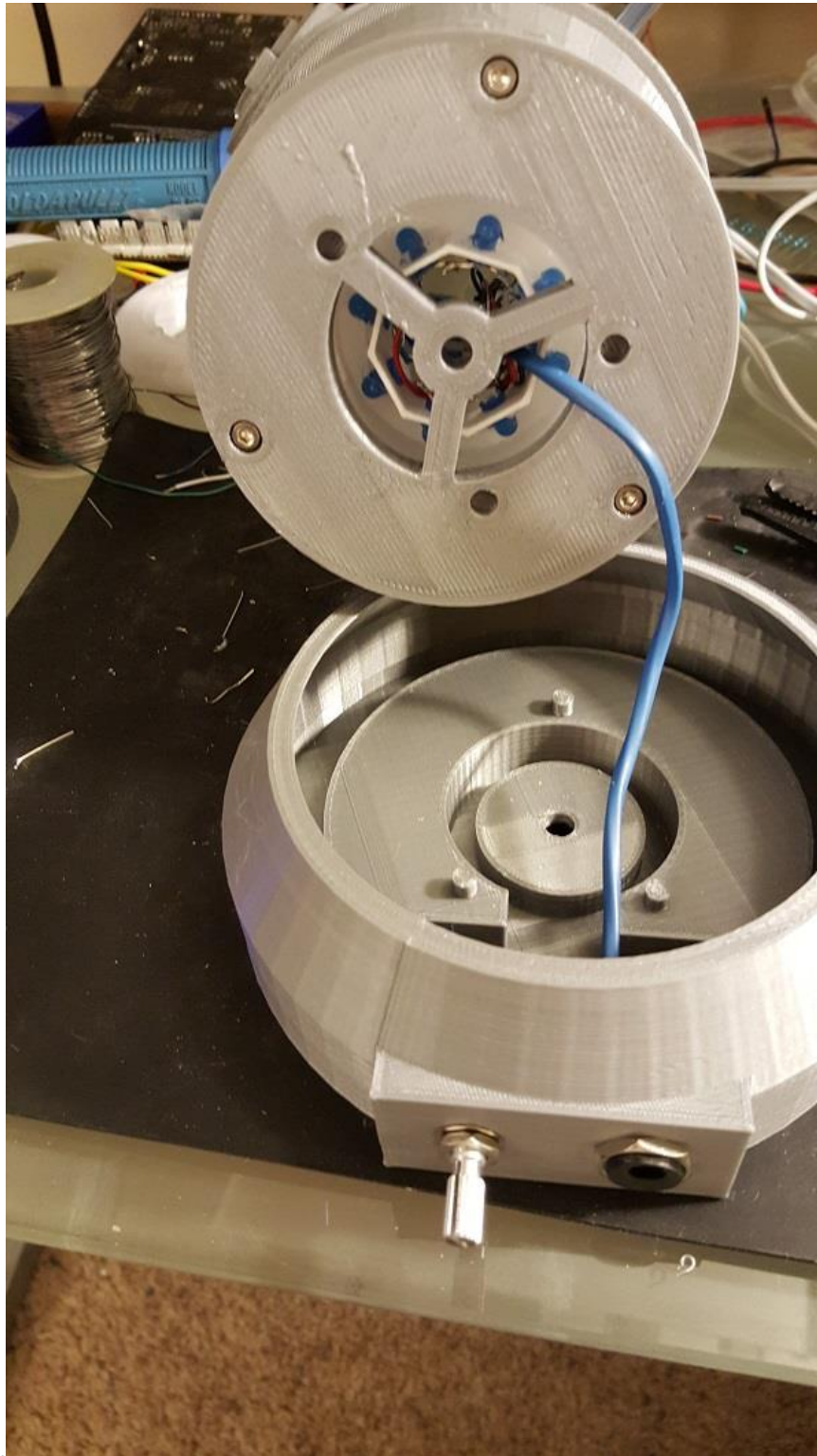
Assemble the other two sections using the same process using ChamberEnd, SupportRingToRingLower and SupportRingToRingUpper, SupportRingToMI.

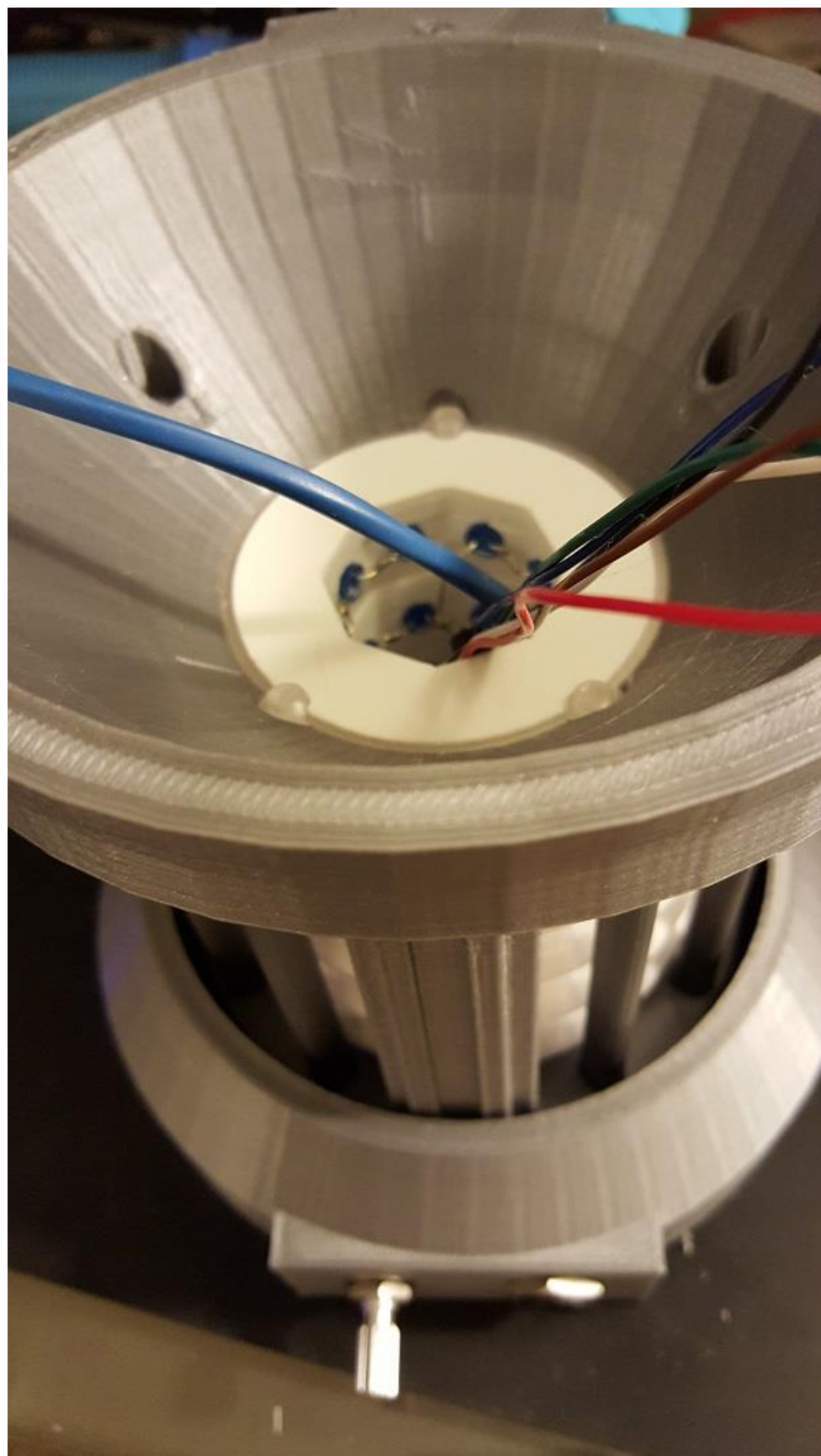
Set the three light segments aside and assemble the WarpCoreBase.



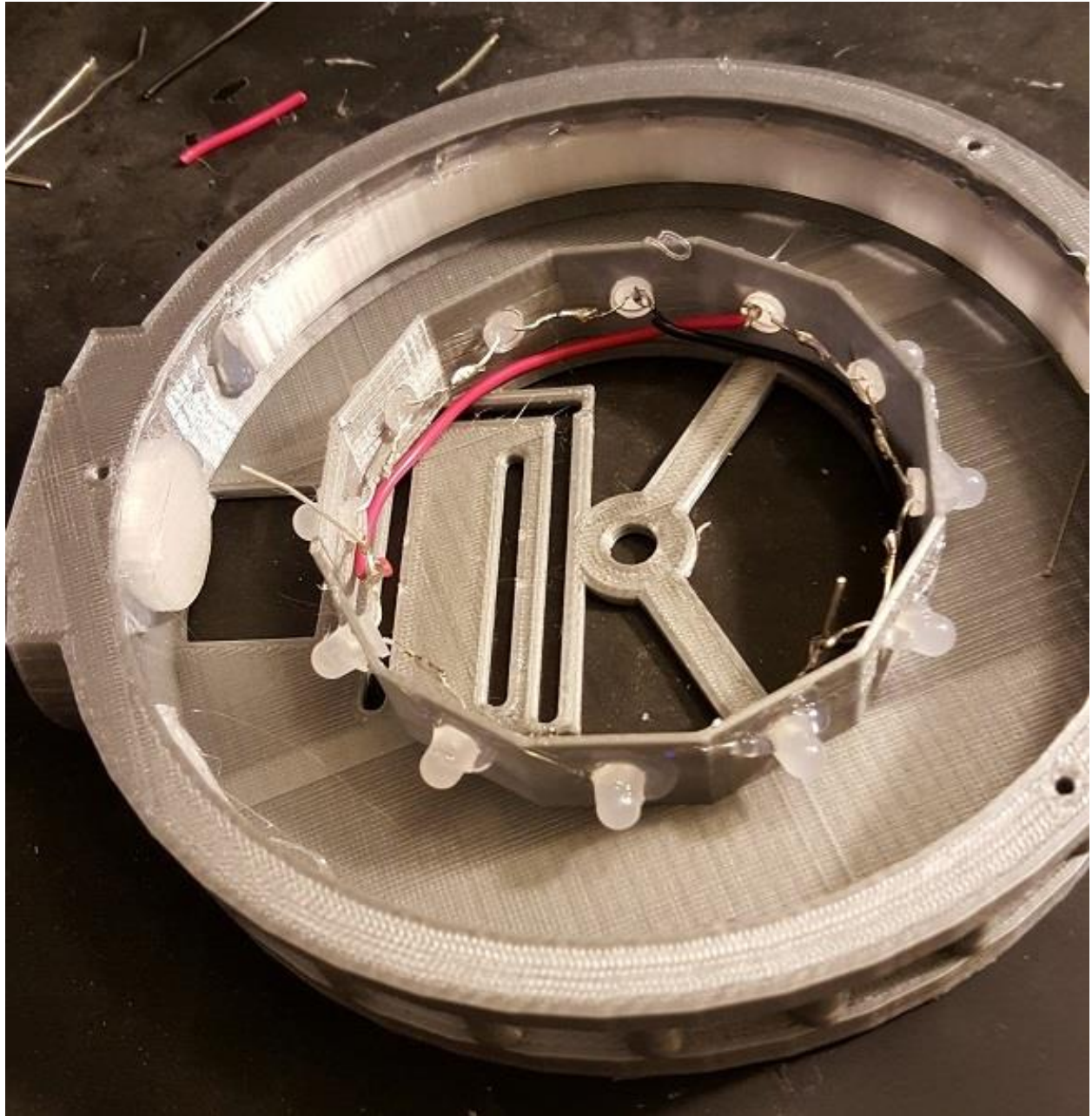
I found getting the wire lengths right was easy if I used the mounting holes in the base as shown above. The red wire attaches to the center connector of the power plug. This is +12vdc. The black wire goes to the outer connector, or ground. The top connector on the pot is connected to ground. White to the center and green to the bottom. I then moved the assembly to the inside and bolted everything into place

The four wires are routed up through the light unit made with the SupportRingToBase (there is no tab on this one like there is on the other two). The light unit can then be set into place on the base, aligning the nubs into the matching holes. They only fit one way.

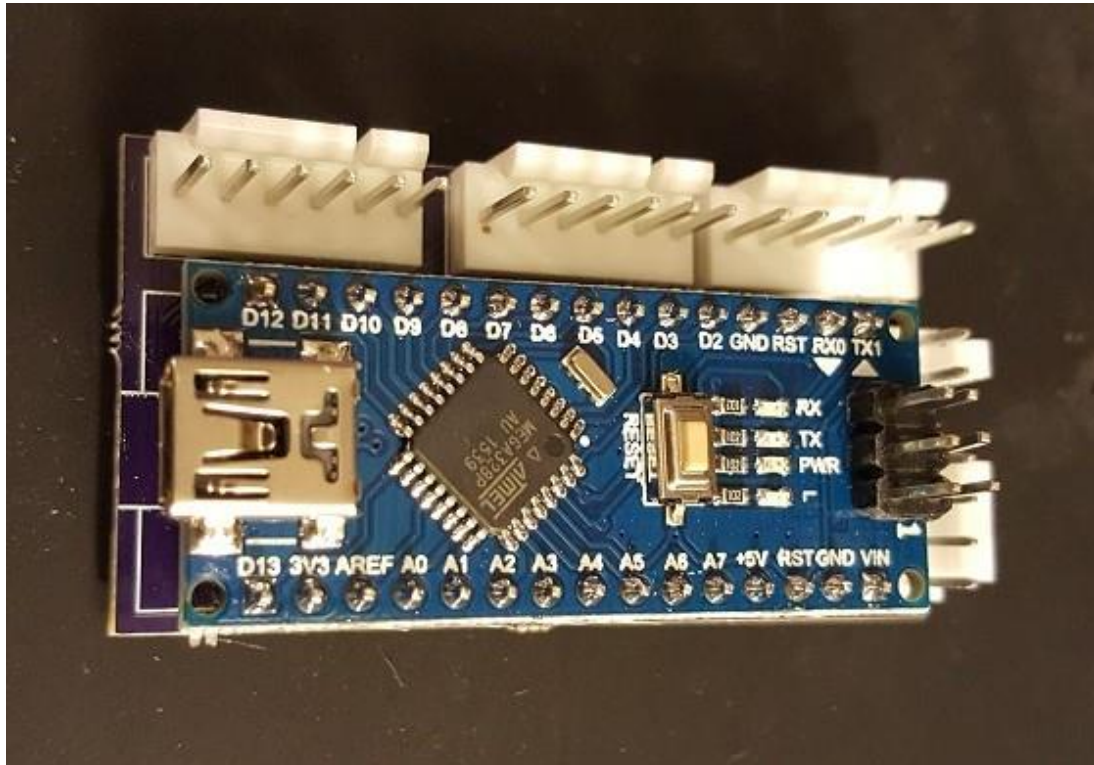
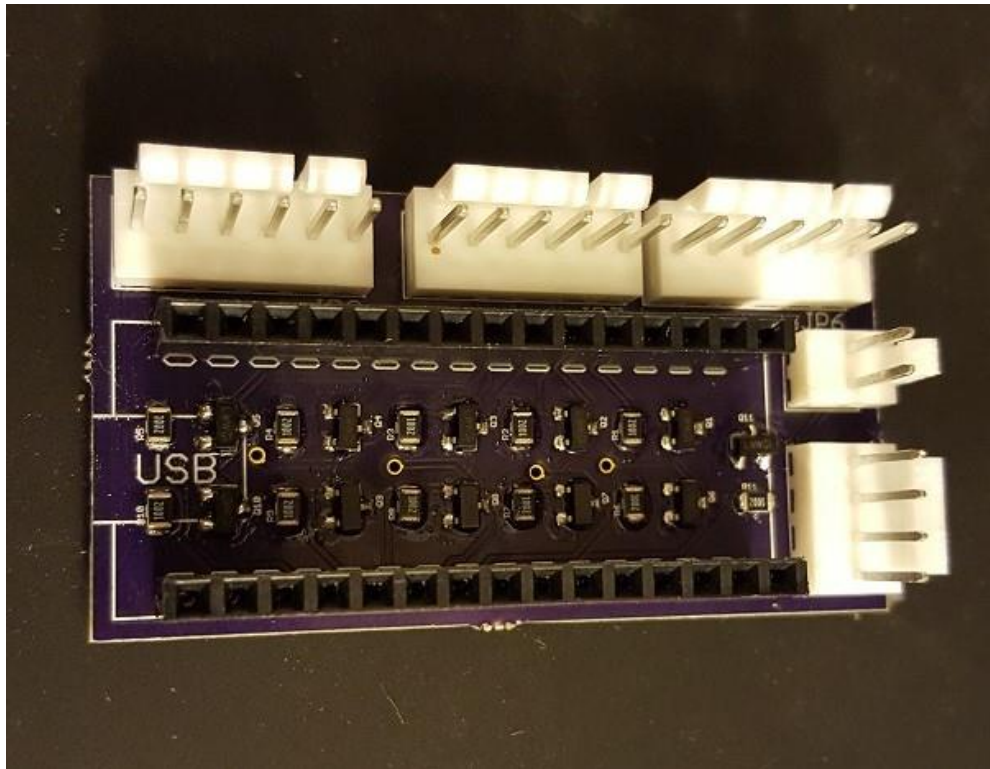


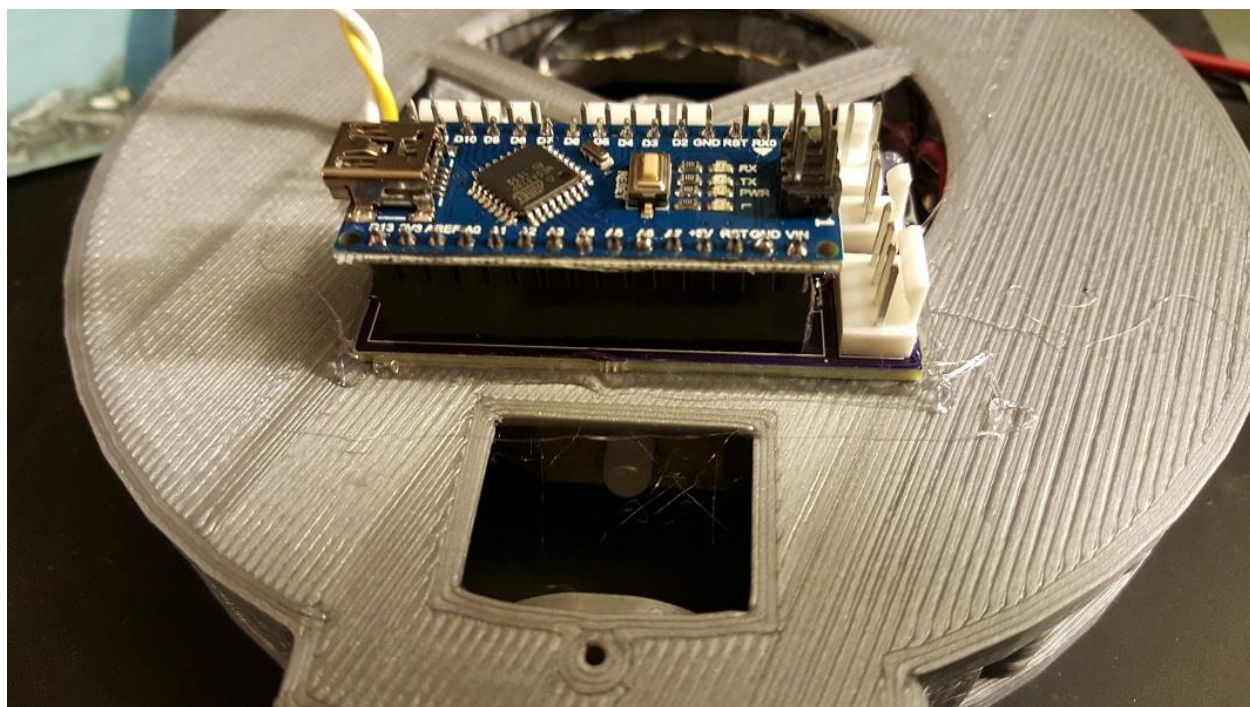


Using a similar technic as the light shelves, assemble the LEDS in the ChamberMiddle. There are 3 sets of 4 LEDS in series in this section. You will need to run a couple extra wires to connect the anode and cathode of the third set of LEDS to the other two. The lead sticking up at the 9 o'clock position is the anode and the lead at 4 o'clock is the cathode.

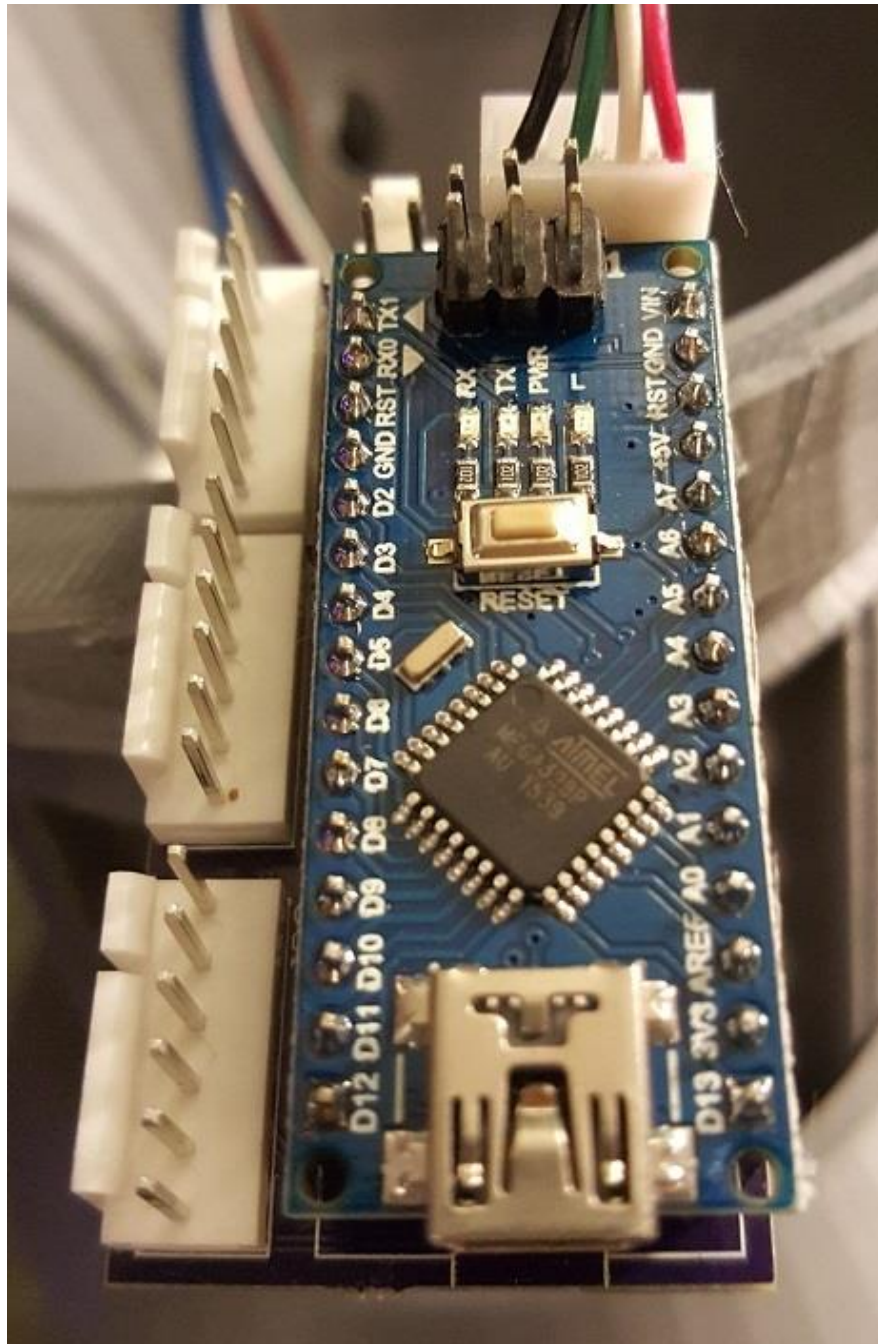


Next, assemble the Arduino Nano and attach to the bottom of the ChamberMiddle unit.

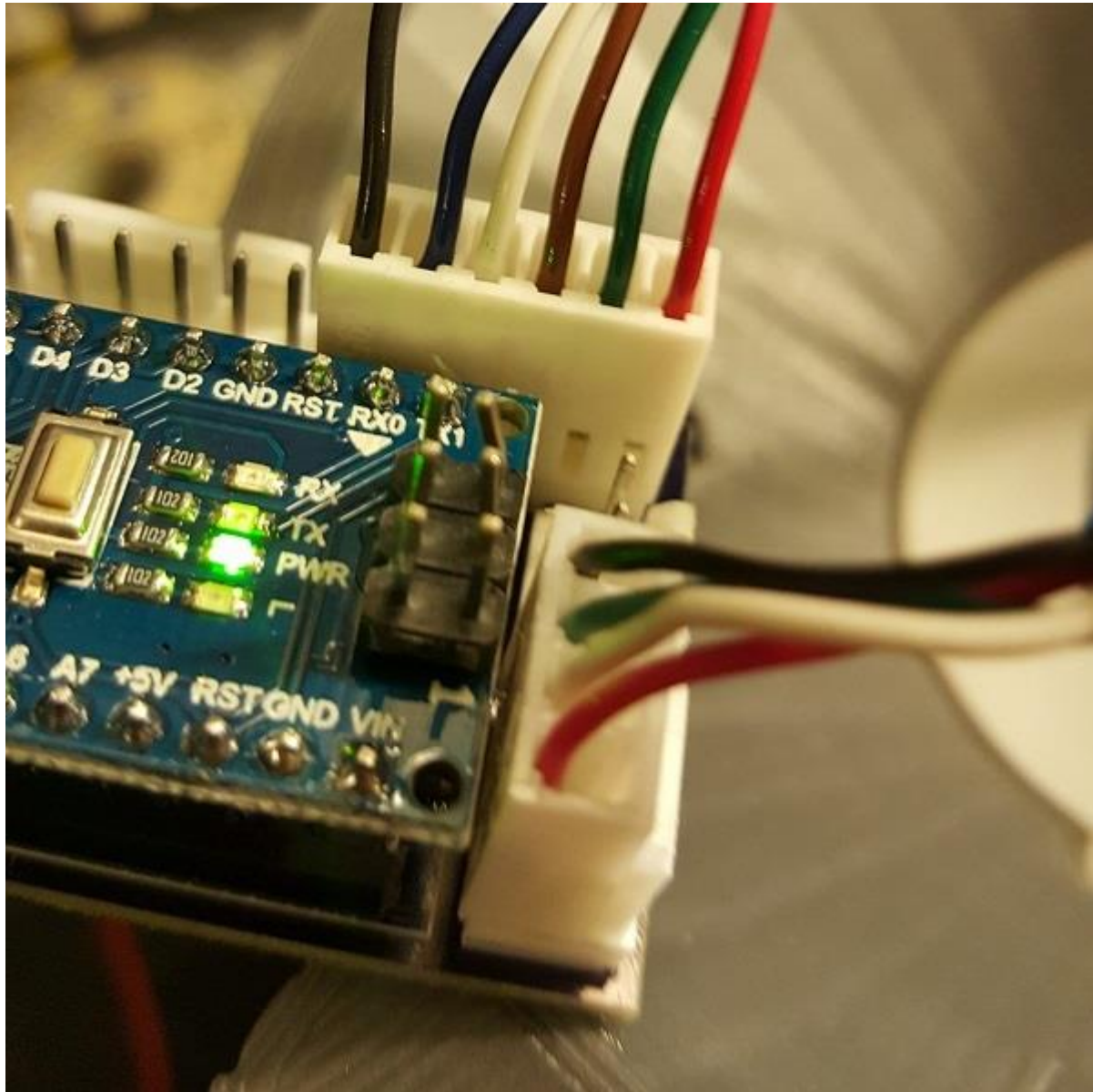




Wire the four conductor cable from the power/pot as shown below. Black is ground. Green goes to the bottom connection of the pot. White to the middle connector. Red to +12vdc.



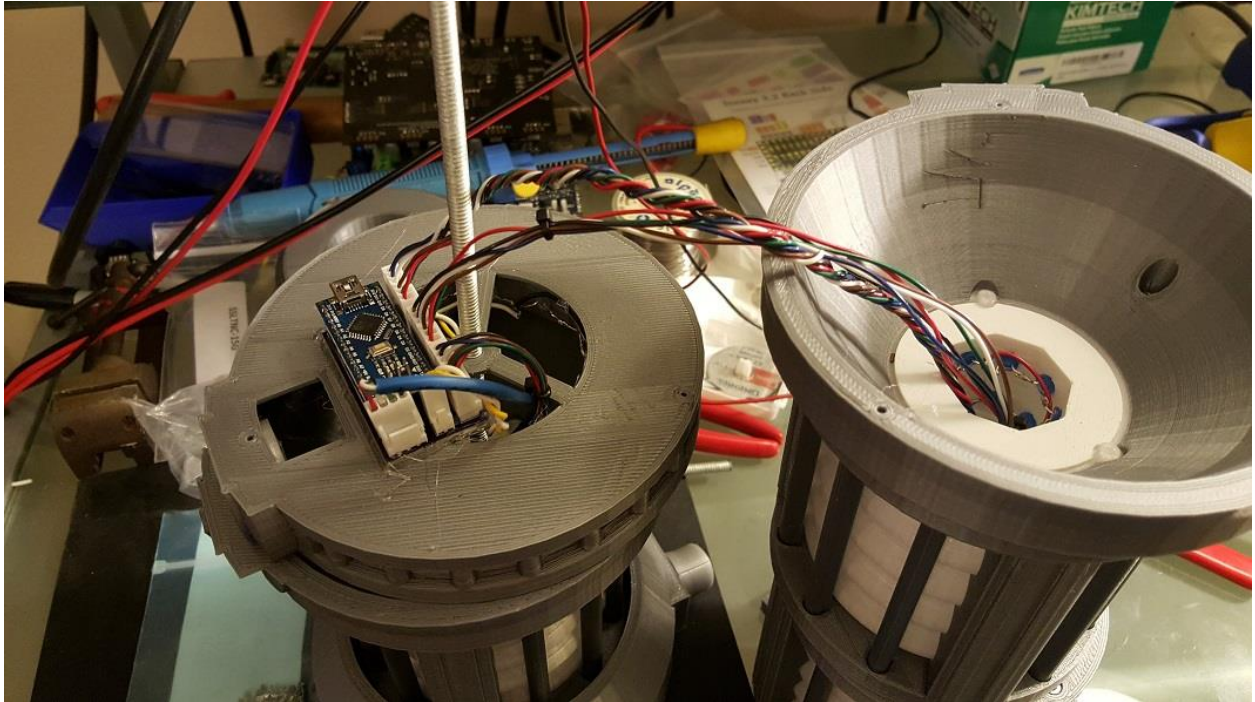
Wire up the three light sections. Connect a six pin molex header to the six wires from each section. If you used the same wire colors as I did, this is how it will look. Each group is wired the same. The right two connectors go to the two light sections around the chamber. It doesn't matter which plug is used.



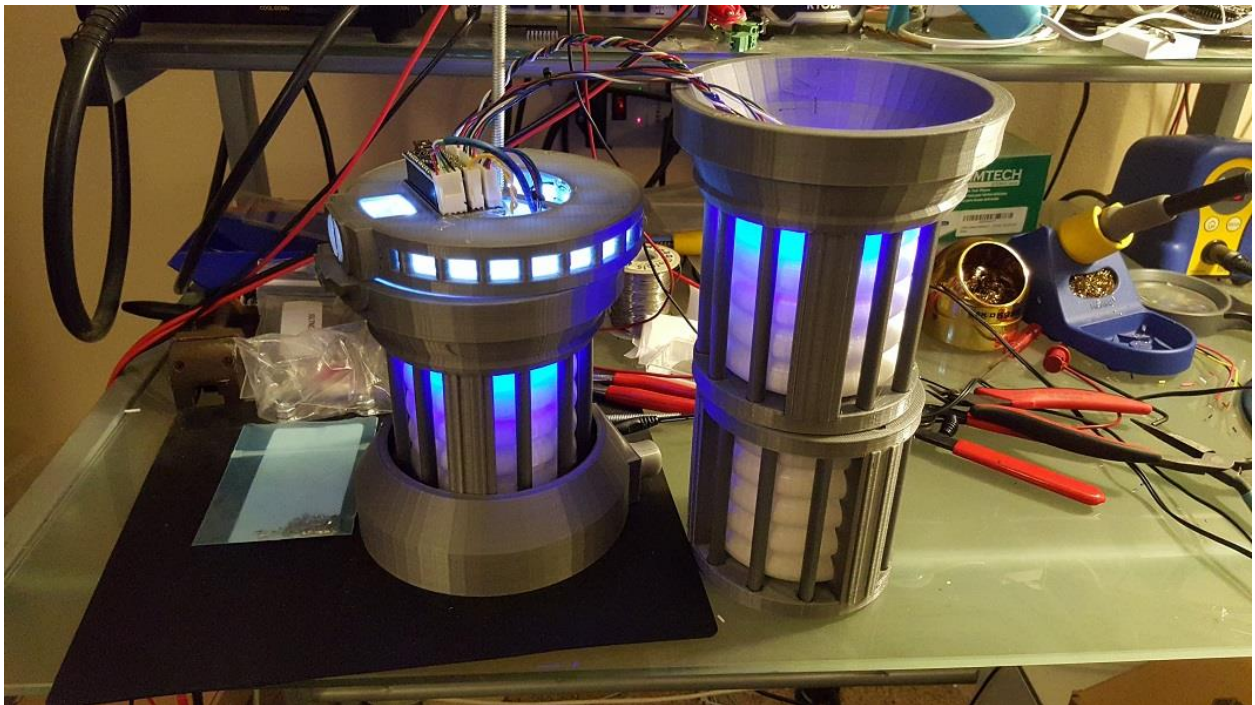
Wire the two pin connector to the chamber lights. The bottom pin connects to the anode. Top pin to the cathode.

Now would probably be a good time to glue in the Warp Core Dilithium Plug and MARC Lights Extended pieces.

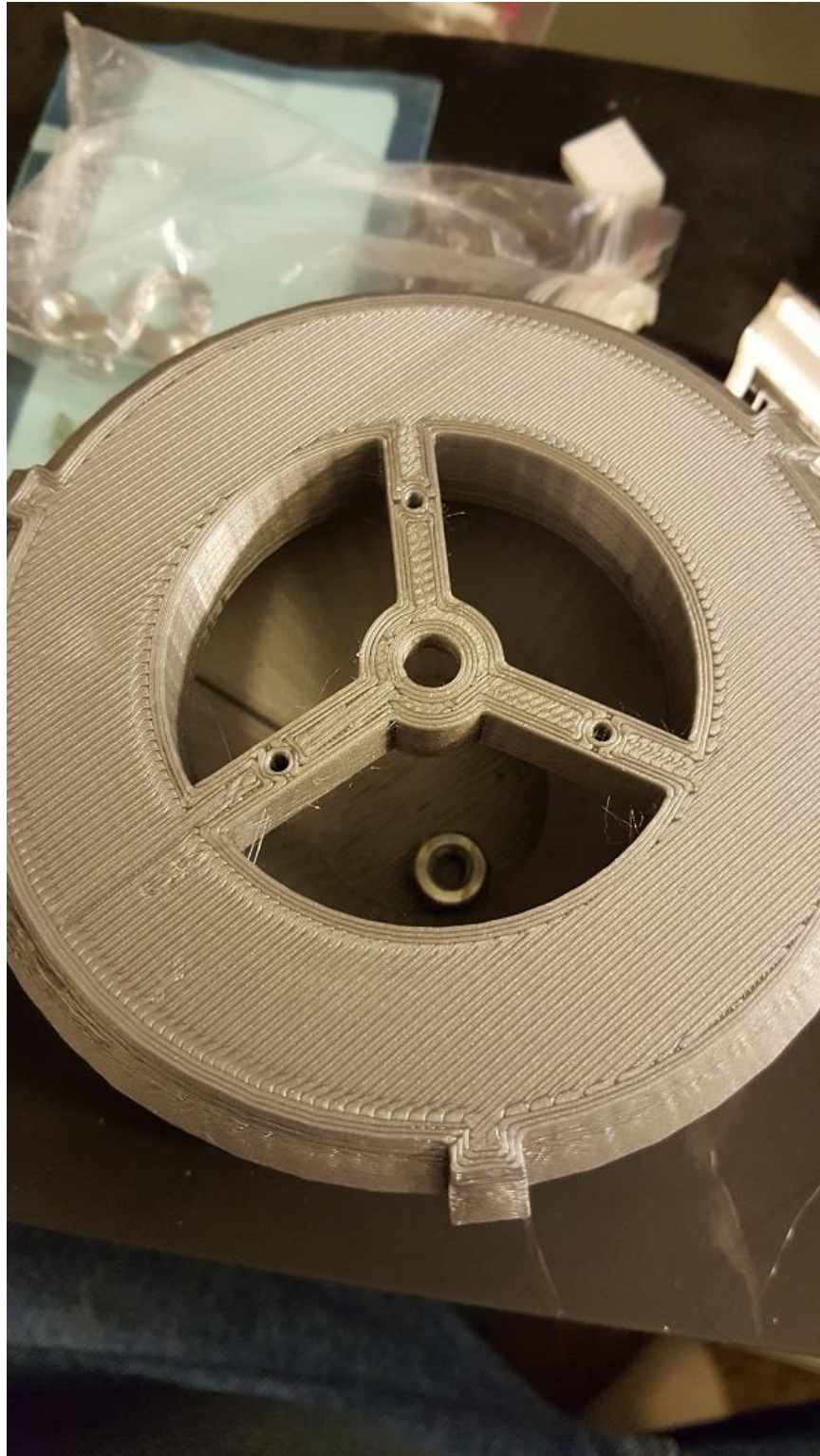
To wire up the top two sections, feed the wires from the top most section (one with the SupportRingToMI) through the center of the other section. It can help if the wires are twisted together or otherwise bundled. Connect each section to the Nano.



At this point, you can power up the unit. If everything is wired and working correctly, you should see the lights cascade in sequence. The lights in the chamber will pulse when the cascade reaches it.

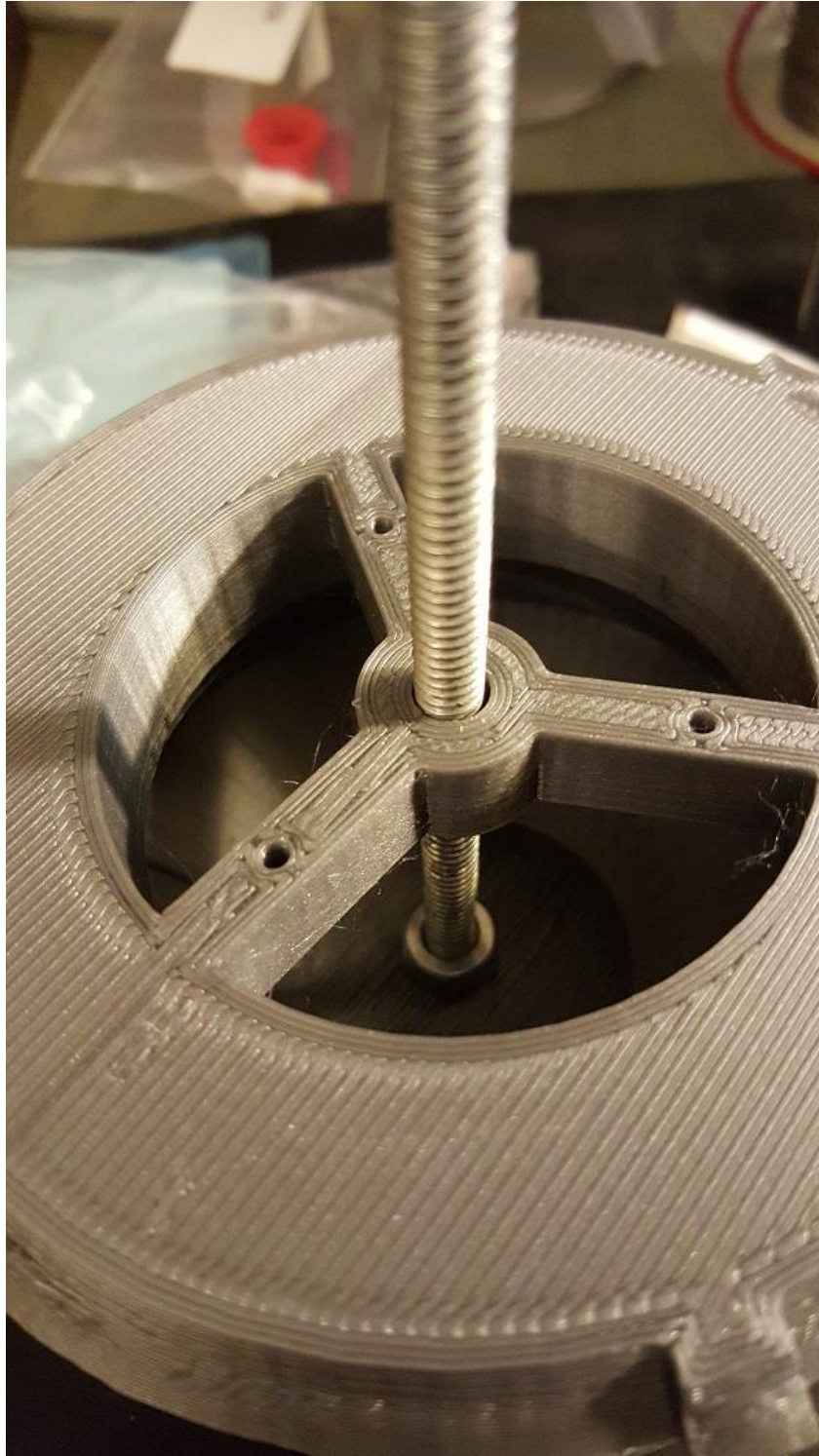


Embed one of the $\frac{1}{4}$ " nuts into the Matter Injector. I found it easiest to do by first dropping the nut into the Matter Injector...



Place a $\frac{1}{4}$ " nut and washer on the rod allowing plenty of space for the rod to reach the nut in the Matter Injector.

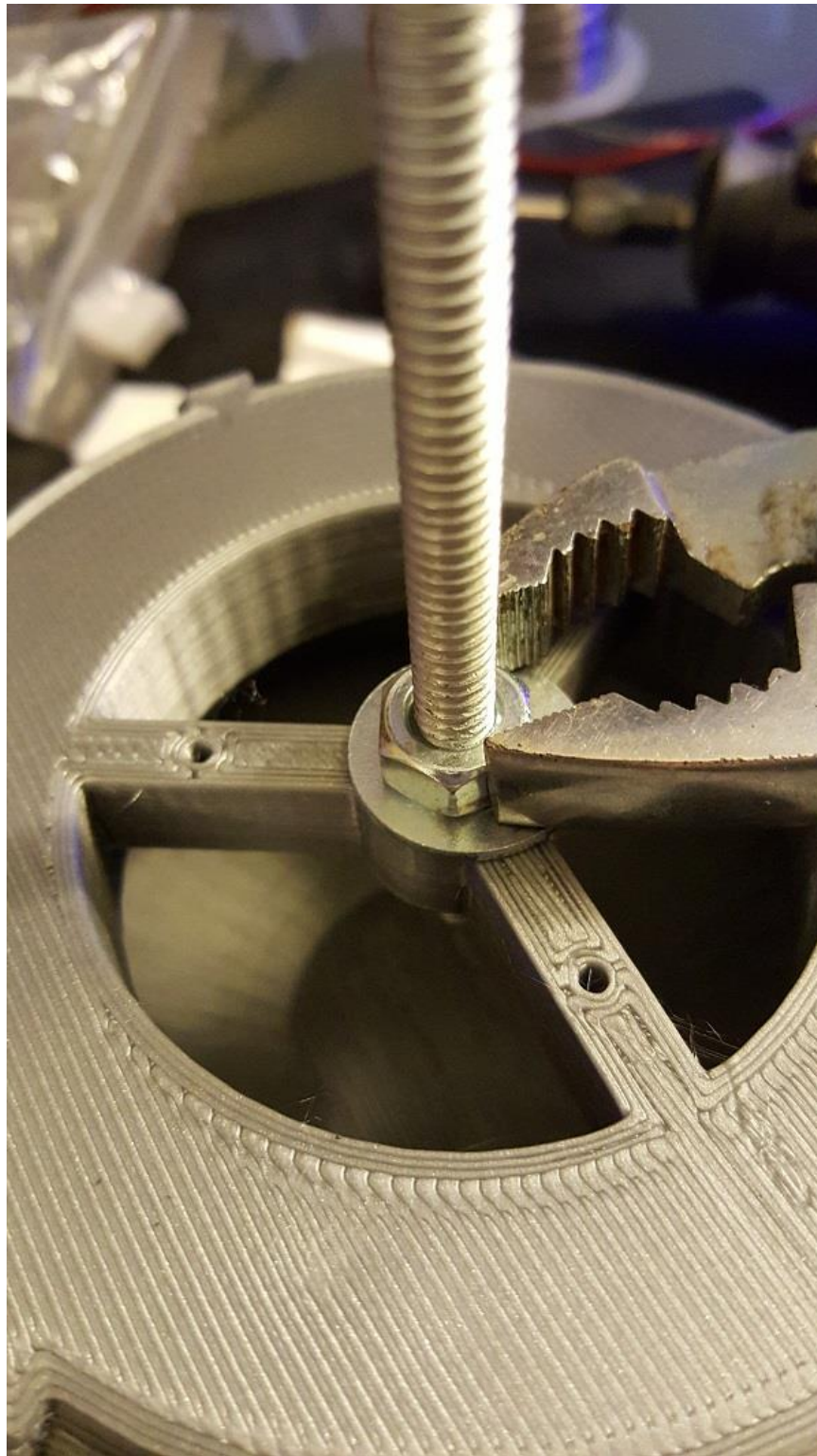
Place the $\frac{1}{4}$ " threaded rod through the hole and onto the top of the nut. Reach down with your fingers or other long object to hold the nut in place and thread the rod into the nut.



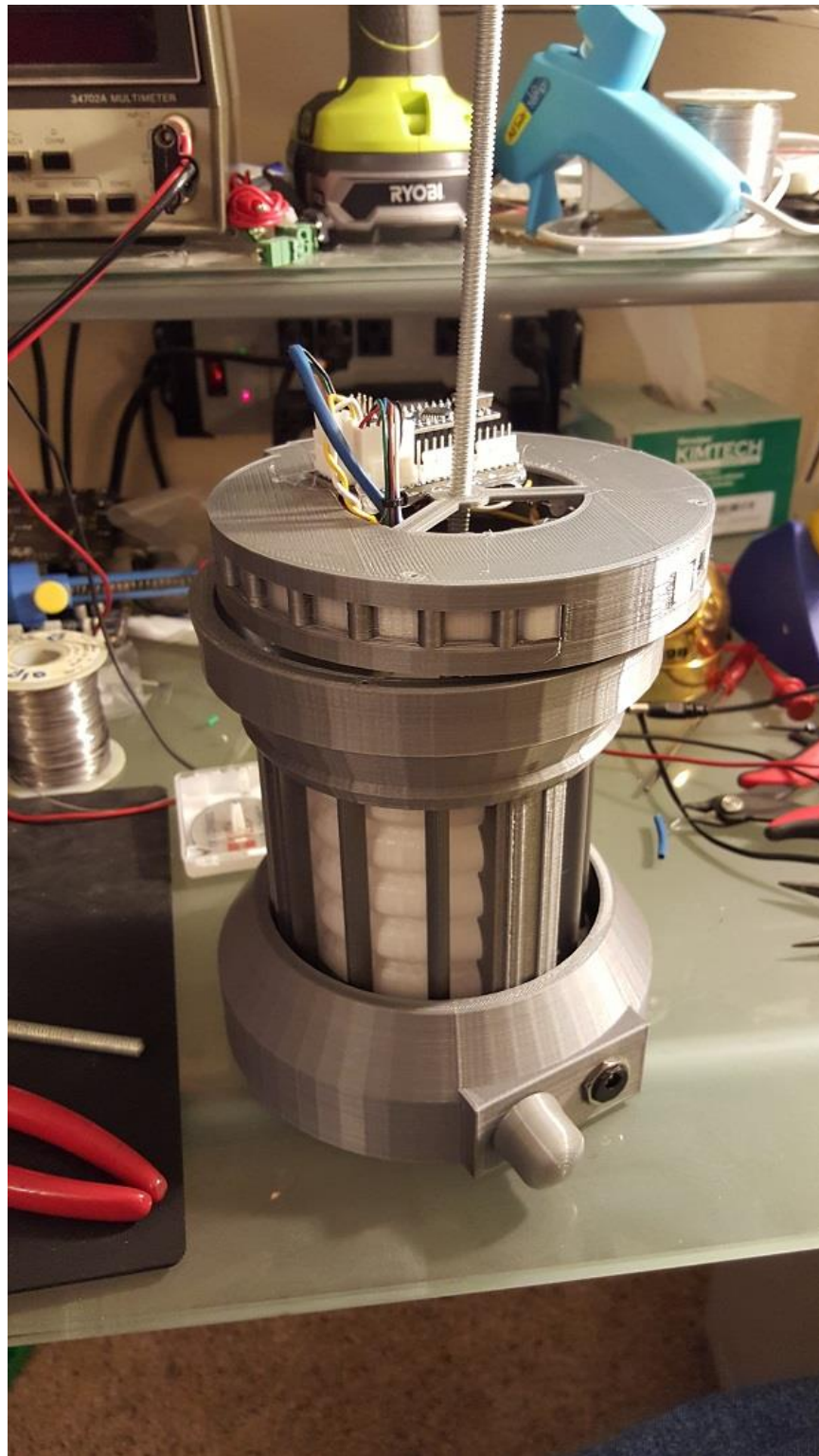
Pull the rod with the nut up and tighten the second nut and washer.



Keep tightening the outer nut until the inner nut seats itself and then remove the rod.



Move the nut and washer to one of the ends of the rod. Slide the rod up through the Warp Core base, lower light section, and chamber middle.



Continue sliding the rod through the upper two light sections. You will have to pull the rod out to be able to thread the rod through the upper light sections .



The following step is optional, but I find it helpful to keep the sections aligned. Where each of the pieces sit on each other, other than the base to the first light section, there are small holes. They holes are sized such that a piece of 1.75mm filament will fit in them. There are matching holes on the opposite piece for the filament pieces to fit into. Each section will only fit together one way using these alignment holes.

Once all the sections are fitted together, screw the Matter Injector cap onto the top to tighten everything together. Depending on how long the threaded rod is, you may have to cut it so it will fit.

Plug in the Warp Core and enjoy.



If you find that I have left out something, or things are not clear, please message me on Thingiverse. I'll do my best to help clear up any issues.